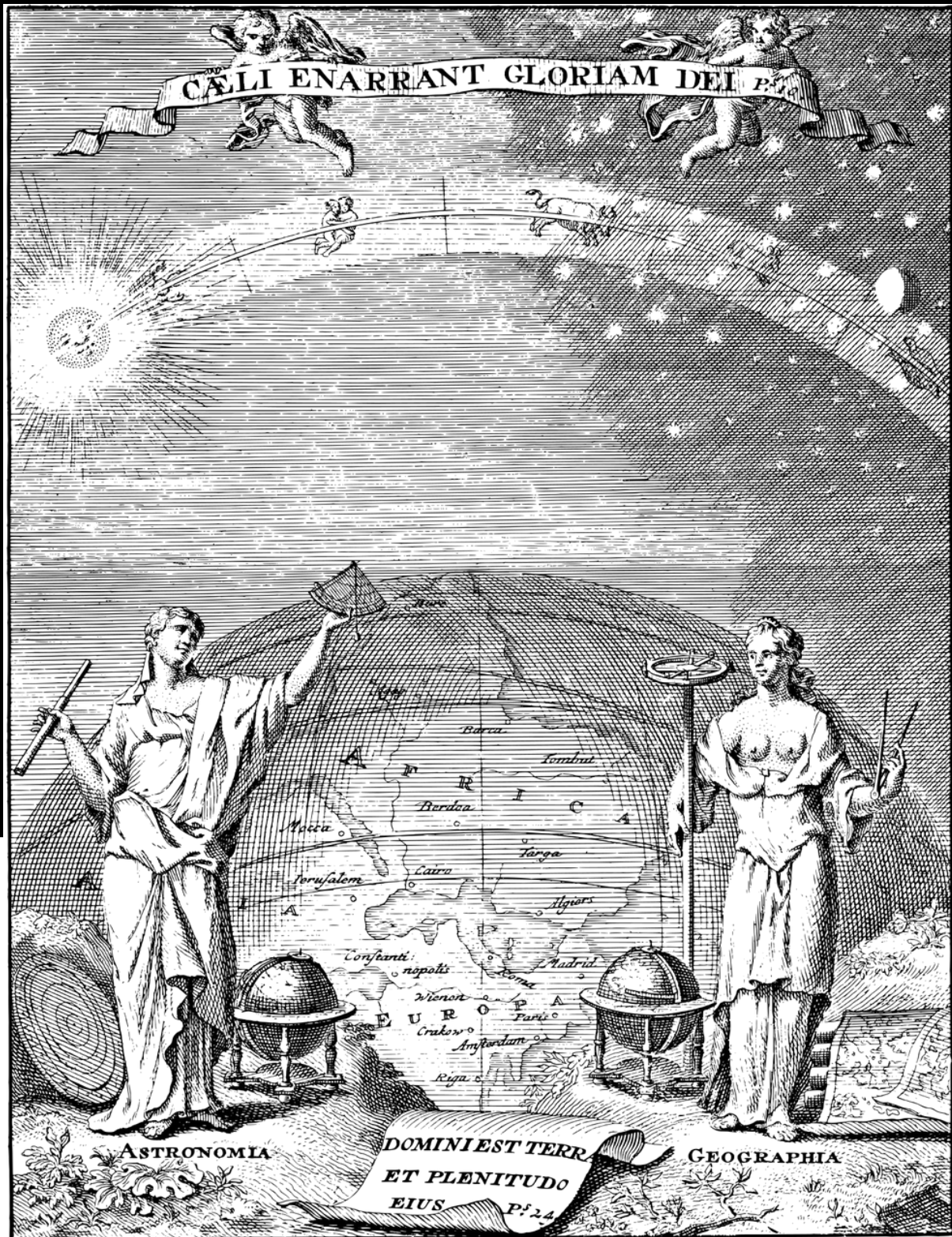


Introduction to Academic Research



Edited by:
Sander van Lanen

Foreword

This is the textbook for the course Introduction to Academic Research, taught at the Faculty of Spatial Sciences, University of Groningen. Although my name is on the cover, I did not write more than a few words of the chapters of this book. I am not the author, but rather the compiler of this book.

The content of the four chapters comes from different textbooks that are classified as 'open educational resources' (OER). OER are texts and other media that can be retained, revised, remixed, reused and redistributed. This material is thus freely useable and shareable. For this textbook, OER allowed me to compile four chapters from three different sources that together provide the accompanying literature for the course Introduction to Academic Research.

The original sources of the chapters in this book are mentioned on every page. These texts are taken almost verbatim from these sources, I only made small adjustments for the consistency of referencing, providing a little context, or aligning the numbering of sections. Because I build so much on their texts, I am heavily indebted to Heather Griffiths, Nathan Keirns, Steven Graves, and Anol Bhattacharjee for making their resources openly available. I also want to thank Myroslava Zhuk, Frederiek van Rij, and Raquel Raj for their information, encouragement and advice regarding the use of Open Educational Resources.

This is the first version of this textbook. Because OER allows for the revising and remising of material, the content and order of the texts in this book might be adjusted to better suit the needs of Introduction to Academic Research in the future. Such improvements will be based on my experiences as a lecturer and student feedback, so please don't hesitate to contact me with any suggestions, questions, or comments about the content, order, or layout in this book.

I hope you will enjoy this collection, appreciate open learning, and will learn about the practice, history and application of research in the spatial sciences.

Sander van Lanen
Groningen, 2021

Cover image by Gordon Johnson from [Pixabay](https://pixabay.com/)

Chapter 1: Academic research and the research process

Have you ever wondered if home schooling affects a person's later success in college or how many people wait until they are in their forties to get married? Do you wonder if texting is changing teenagers' abilities to spell correctly or to communicate clearly? How do social movements like Occupy Wall Street develop? How about the development of social phenomena like the massive public followings for Star Trek and Harry Potter? The goal of research is to answer questions. Sociological research attempts to answer a vast variety of questions, such as these and more, about our social world.



Figure 1.1 Many believe that crime rates go up during the full moon, but scientific research does not support this conclusion. (Photo courtesy of Jubula 2/flickr)

We often have opinions about social situations, but these may be biased by our expectations or based on limited data. Instead, scientific research is based on **empirical evidence**, which is evidence that comes from direct experience, scientifically gathered data, or experimentation. Many people believe, for example, that crime rates go up when there's a full moon, but research doesn't support this opinion. Researchers Rotton and Kelly (1985) conducted a **meta-analysis** of research on the full moon's effects on behavior. Meta-analysis is a technique in which the results of virtually all previous studies on a specific subject are evaluated together. Rotton and Kelly's meta-analysis included thirty-seven prior studies on the effects of the full moon on crime rates, and the overall findings were that full moons are entirely unrelated to crime, suicide, psychiatric problems, and crisis center calls (cited in Arkowitz and Lilienfeld 2009). We may each know of an instance in which a crime happened during a full moon, but it was likely just a coincidence.

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/1-introduction-to-sociology>

People commonly try to understand the happenings in their world by finding or creating an explanation for an occurrence. Social scientists may develop a **hypothesis** for the same reason. A hypothesis is a testable educated guess about predicted outcomes between two or more variables; it's a possible explanation for specific happenings in the social world and allows for testing to determine whether the explanation holds true in many instances, as well as among various groups or in different places. Sociologists use empirical data and the **scientific method**, or an interpretative framework, to increase understanding of societies and social interactions, but research begins with the search for an answer to a question.

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/1-introduction-to-sociology>

1.1 Approaches to sociological research

When sociologists apply the sociological perspective and begin to ask questions, no topic is off limits. Every aspect of human behavior is a source of possible investigation. Sociologists question the world that humans have created and live in. They notice patterns of behavior as people move through that world. Using sociological methods and systematic research within the framework of the scientific method and a scholarly interpretive perspective, sociologists have discovered workplace patterns that have transformed industries, family patterns that have enlightened family members, and education patterns that have aided structural changes in classrooms.

The crime during a full moon discussion put forth a few loosely stated opinions. If the human behaviors around those claims were tested systematically, a police officer, for example, could write a report and offer the findings to sociologists and the world in general. The new perspective could help people understand themselves and their neighbors and help people make better decisions about their lives. It might seem strange to use scientific practices to study social trends, but, as we shall see, it's extremely helpful to rely on systematic approaches that research methods provide.

Sociologists often begin the research process by asking a question about how or why things happen in this world. It might be a unique question about a new trend or an old question about a common aspect of life. Once the sociologist forms the question, he or she proceeds through an in-depth process to answer it. In deciding how to design that process, the researcher may adopt a scientific approach or an interpretive framework. The following sections describe these approaches to knowledge.

The Scientific Method

Sociologists make use of tried and true methods of research, such as experiments, surveys, and field research. But humans and their social interactions are so diverse that these interactions can seem impossible to chart or explain. It might seem that science is about discoveries and chemical reactions or about proving ideas right or wrong rather than about exploring the nuances of human behavior.

However, this is exactly why scientific models work for studying human behavior. A scientific process of research establishes parameters that help make sure results are objective and accurate. Scientific methods provide limitations and boundaries that focus a study and organize its results.

The scientific method involves developing and testing theories about the world based on empirical evidence. It is defined by its commitment to systematic observation of the empirical world and strives to be objective, critical, skeptical, and logical. It involves a series of prescribed steps that have been established over centuries of scholarship.

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/2-1-approaches-to-sociological-research>

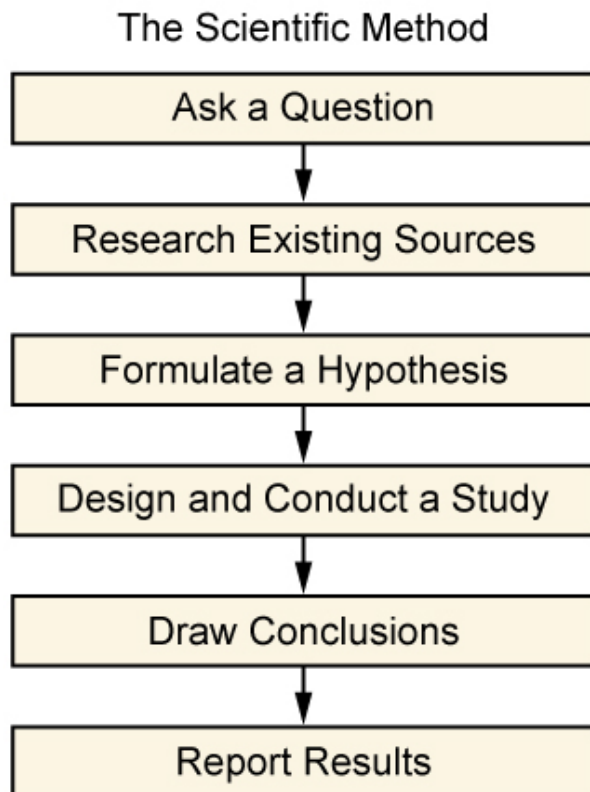


Figure 1.2 The scientific method is an essential tool in research.

But just because sociological studies use scientific methods does not make the results less human. Sociological topics are not reduced to right or wrong facts. In this field, results of studies tend to provide people with access to knowledge they did not have before—knowledge of other cultures, knowledge of rituals and beliefs, or knowledge of trends and attitudes. No matter what research approach they use, researchers want to maximize the study's **reliability**, which refers to how likely research results are to be replicated if the study is reproduced. Reliability increases the likelihood that what happens to one person will happen to all people in a group. Researchers also strive for **validity**, which refers to how well the study measures what it was designed to measure. Returning to the crime rate during a full moon topic, reliability of a study would reflect how well the resulting experience represents the average adult crime rate during a full moon. Validity would ensure that the study's design accurately examined what it was designed to study, so an exploration of adult criminal behaviors during a full moon should address that issue and not veer into other age groups' crimes, for example.

In general, sociologists tackle questions about the role of social characteristics in outcomes. For example, how do different communities fare in terms of psychological well-being, community cohesiveness, range of vocation, wealth, crime rates, and so on? Are communities functioning smoothly? Sociologists look between the cracks to discover obstacles to meeting basic human needs. They might study environmental influences and patterns of behavior that lead to crime, substance abuse, divorce, poverty, unplanned pregnancies, or illness. And, because sociological studies are not all focused on negative behaviors or challenging

situations, researchers might study vacation trends, healthy eating habits, neighborhood organizations, higher education patterns, games, parks, and exercise habits.

Sociologists can use the scientific method not only to collect but also to interpret and analyze the data. They deliberately apply scientific logic and objectivity. They are interested in—but not attached to—the results. They work outside of their own political or social agendas. This doesn't mean researchers do not have their own personalities, complete with preferences and opinions. But sociologists deliberately use the scientific method to maintain as much objectivity, focus, and consistency as possible in a particular study.

With its systematic approach, the scientific method has proven useful in shaping sociological studies. The scientific method provides a systematic, organized series of steps that help ensure objectivity and consistency in exploring a social problem. They provide the means for accuracy, reliability, and validity. In the end, the scientific method provides a shared basis for discussion and analysis (Merton, 1949).

Typically, the scientific method starts with these steps—1) ask a question, 2) research existing sources, 3) formulate a hypothesis—described below.

Ask a Question

The first step of the scientific method is to ask a question, describe a problem, and identify the specific area of interest. The topic should be narrow enough to study within a geography and time frame. “Are societies capable of sustained happiness?” would be too vague. The question should also be broad enough to have universal merit. “What do personal hygiene habits reveal about the values of students at XYZ High School?” would be too narrow. That said, happiness and hygiene are worthy topics to study. Sociologists do not rule out any topic, but would strive to frame these questions in better research terms.

That is why sociologists are careful to define their terms. In a hygiene study, for instance, hygiene could be defined as “personal habits to maintain physical appearance (as opposed to health),” and a researcher might ask, “How do differing personal hygiene habits reflect the cultural value placed on appearance?” When forming these basic research questions, sociologists develop an **operational definition**, that is, they define the concept in terms of the physical or concrete steps it takes to objectively measure it. The operational definition identifies an observable condition of the concept. By operationalizing a variable of the concept, all researchers can collect data in a systematic or replicable manner.

The operational definition must be valid, appropriate, and meaningful. And it must be reliable, meaning that results will be close to uniform when tested on more than one person. For example, “good drivers” might be defined in many ways: those who use their turn signals, those who don't speed, or those who courteously allow others to merge. But these driving behaviors could be interpreted differently by different researchers and could be difficult to measure. Alternatively, “a driver who has never received a traffic violation” is a specific

description that will lead researchers to obtain the same information, so it is an effective operational definition.

Research Existing Sources

The next step researchers undertake is to conduct background research through a **literature review**, which is a review of any existing similar or related studies. A visit to the library and a thorough online search will uncover existing research about the topic of study. This step helps researchers gain a broad understanding of work previously conducted on the topic at hand and enables them to position their own research to build on prior knowledge. Researchers—including student researchers—are responsible for correctly citing existing sources they use in a study or that inform their work. While it is fine to borrow previously published material (as long as it enhances a unique viewpoint), it must be referenced properly and never plagiarized.

To study hygiene and its value in a particular society, a researcher might sort through existing research and unearth studies about child-rearing, vanity, obsessive-compulsive behaviors, and cultural attitudes toward beauty. It's important to sift through this information and determine what is relevant. Using existing sources educates researchers and helps refine and improve studies' designs.

Formulate a Hypothesis

A **hypothesis** is an assumption about how two or more variables are related; it makes a conjectural statement about the relationship between those variables. In sociology, the hypothesis will often predict how one form of human behavior influences another. In research, **independent variables** are the *cause* of the change. The **dependent variable** is the *effect*, or thing that is changed.

Table 1.1 Examples of Dependent and Independent Variables Typically, the independent variable causes the dependent variable to change in some way.

Hypothesis	Independent Variable	Dependent Variable
The greater the availability of affordable housing, the lower the homeless rate.	Affordable Housing	Homeless Rate
The greater the availability of math tutoring, the higher the math grades.	Math Tutoring	Math Grades
The greater the police patrol presence, the safer the neighbourhood.	Police Patrol Presence	Safer Neighbourhood
The greater the factory lighting, the higher the productivity.	Factory Lighting	Productivity

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/2-1-approaches-to-sociological-research>

The greater the amount of observation, the higher the public awareness.	Observation	Public Awareness
---	-------------	------------------

For example, in a basic study, the researcher would establish one form of human behavior as the independent variable and observe the influence it has on a dependent variable. How does gender (the independent variable) affect the rate of income (the dependent variable)? How does one's religion (the independent variable) affect family size (the dependent variable)? How is social class (the dependent variable) affected by level of education (the independent variable)?

At this point, a researcher's operational definitions help measure the variables. In a study asking how tutoring improves grades, for instance, one researcher might define a "good" grade as a C or better, while another uses a B+ as a starting point for "good." Another operational definition might describe "tutoring" as "one-on-one assistance by an expert in the field, hired by an educational institution." Those definitions set limits and establish cut-off points that ensure consistency and replicability in a study.

As the table shows, an independent variable is the one that causes a dependent variable to change. For example, a researcher might hypothesize that teaching children proper hygiene (the independent variable) will boost their sense of self-esteem (the dependent variable). Or rephrased, a child's sense of self-esteem depends, in part, on the quality and availability of hygienic resources.

Of course, this hypothesis can also work the other way around. Perhaps a sociologist believes that increasing a child's sense of self-esteem (the independent variable) will automatically increase or improve habits of hygiene (now the dependent variable). Identifying the independent and dependent variables is very important. As the hygiene example shows, simply identifying two topics, or variables, is not enough; their prospective relationship must be part of the hypothesis.

Just because a sociologist forms an educated prediction of a study's outcome doesn't mean data contradicting the hypothesis aren't welcome. Sociologists analyze general patterns in response to a study, but they are equally interested in exceptions to patterns. In a study of education, a researcher might predict that high school dropouts have a hard time finding rewarding careers. While it has become at least a cultural assumption that the higher the education, the higher the salary and degree of career happiness, there are certainly exceptions. People with little education have had stunning careers, and people with advanced degrees have had trouble finding work. A sociologist prepares a hypothesis knowing that results will vary.

Once the preliminary work is done, it's time for the next research steps: designing and conducting a study and drawing conclusions. These research methods are discussed below.

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/2-1-approaches-to-sociological-research>

Interpretive Framework

While many sociologists rely on the scientific method as a research approach, others operate from an ***interpretive framework***. While systematic, this approach doesn't follow the hypothesis-testing model that seeks to find generalizable results. Instead, an *interpretive framework*, sometimes referred to as an interpretive perspective, seeks to understand social worlds from the point of view of participants, which leads to in-depth knowledge.

Interpretive research is generally more descriptive or narrative in its findings. Rather than formulating a hypothesis and method for testing it, an interpretive researcher will develop approaches to explore the topic at hand that may involve a significant amount of direct observation or interaction with subjects. This type of researcher also learns as he or she proceeds and sometimes adjusts the research methods or processes midway to optimize findings as they evolve

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/2-1-approaches-to-sociological-research>

1.2 Research methods

Sociologists examine the world, see a problem or interesting pattern, and set out to study it. They use research methods to design a study—perhaps a detailed, systematic, scientific method for conducting research and obtaining data, or perhaps an ethnographic study utilizing an interpretive framework. Planning the research design is a key step in any sociological study.

When entering a particular social environment, a researcher must be careful. There are times to remain anonymous and times to be overt. There are times to conduct interviews and times to simply observe. Some participants need to be thoroughly informed; others should not know they are being observed. A researcher wouldn't stroll into a crime-ridden neighborhood at midnight, calling out, "Any gang members around?" And if a researcher walked into a coffee shop and told the employees they would be observed as part of a study on work efficiency, the self-conscious, intimidated baristas might not behave naturally. This is called the **Hawthorne effect**—where people change their behavior because they know they are being watched as part of a study. The Hawthorne effect is unavoidable in some research. In many cases, sociologists have to make the purpose of the study known. Subjects must be aware that they are being observed, and a certain amount of artificiality may result (Sonnenfeld 1985).

Making sociologists' presence invisible is not always realistic for other reasons. That option is not available to a researcher studying prison behaviors, early education, or the Ku Klux Klan. Researchers can't just stroll into prisons, kindergarten classrooms, or Klan meetings and unobtrusively observe behaviors. In situations like these, other methods are needed. All studies shape the research design, while research design simultaneously shapes the study. Researchers choose methods that best suit their study topics and that fit with their overall approaches to research.

In planning studies' designs, sociologists generally choose from four widely used methods of social investigation: survey, field research, experiment, and **secondary data analysis**, or use of existing sources. Every research method comes with plusses and minuses, and the topic of study strongly influences which method or methods are put to use.

Surveys

As a research method, a survey collects data from subjects who respond to a series of questions about behaviors and opinions, often in the form of a questionnaire. The survey is one of the most widely used scientific research methods. The standard survey format allows individuals a level of anonymity in which they can express personal ideas.



Figure 1.3 Questionnaires are a common research method; the U.S. Census is a well-known example. (Photo courtesy of Kathryn Decker/flickr)

At some point, most people in the United States respond to some type of survey. The U.S. Census is an excellent example of a large-scale survey intended to gather sociological data. Not all surveys are considered sociological research, however, and many surveys people commonly encounter focus on identifying marketing needs and strategies rather than testing a hypothesis or contributing to social science knowledge. Questions such as, "How many hot dogs do you eat in a month?" or "Were the staff helpful?" are not usually designed as scientific research. Often, polls on television do not reflect a general population, but are merely answers from a specific show's audience. Polls conducted by programs such as *American Idol* or *So You Think You Can Dance* represent the opinions of fans but are not particularly scientific. A good contrast to these are the Nielsen Ratings, which determine the popularity of television programming through scientific market research.

Sociologists conduct surveys under controlled conditions for specific purposes. Surveys gather different types of information from people. While surveys are not great at capturing the ways people really behave in social situations, they are a great method for discovering how people feel and think—or at least how they say they feel and think. Surveys can track preferences for presidential candidates or reported individual behaviors (such as sleeping, driving, or texting habits) or factual information such as employment status, income, and education levels.

A survey targets a specific **population**, people who are the focus of a study, such as college athletes, international students, or teenagers living with type 1 (juvenile-onset) diabetes. Most researchers choose to survey a small sector of the population, or a **sample**: that is, a manageable number of subjects who *represent* a larger population. The success of a study depends on how well a population is represented by the sample. In a **random sample**, every person in a population has the same chance of being chosen for the study. According to the

laws of probability, random samples represent the population as a whole. For instance, a Gallup Poll, if conducted as a nationwide random sampling, should be able to provide an accurate estimate of public opinion whether it contacts 2,000 or 10,000 people.



Figure 1.4 American Idol uses a real-time survey system—with numbers—that allows members in the audience to vote on contestants. (Photo courtesy of Sam Howzit/flickr)

After selecting subjects, the researcher develops a specific plan to ask questions and record responses. It is important to inform subjects of the nature and purpose of the study up front. If they agree to participate, researchers thank subjects and offer them a chance to see the results of the study if they are interested. The researcher presents the subjects with an instrument, which is a means of gathering the information. A common instrument is a questionnaire, in which subjects answer a series of questions. For some topics, the researcher might ask yes-or-no or multiple-choice questions, allowing subjects to choose possible responses to each question. This kind of **quantitative data**—research collected in numerical form that can be counted—are easy to tabulate. Just count up the number of “yes” and “no” responses or correct answers, and chart them into percentages.

Questionnaires can also ask more complex questions with more complex answers—beyond “yes,” “no,” or the option next to a checkbox. In those cases, the answers are subjective and vary from person to person. *How do plan to use your college education? Why do you follow Jimmy Buffett around the country and attend every concert?* Those types of questions require short essay responses, and participants willing to take the time to write those answers will convey personal information about religious beliefs, political views, and morals. Some topics that reflect internal thought are impossible to observe directly and are difficult to discuss honestly in a public forum. People are more likely to share honest answers if they can respond to questions anonymously. This type of information is **qualitative data**—results that are subjective and often based on what is seen in a natural setting. Qualitative information is harder to organize and tabulate. The researcher will end up with a wide range of responses, some of which may be surprising. The benefit of written opinions, though, is the wealth of material that they provide.

An **interview** is a one-on-one conversation between the researcher and the subject, and it is a way of conducting surveys on a topic. Interviews are similar to the short-answer questions on surveys in that the researcher asks subjects a series of questions. However, participants are free to respond as they wish, without being limited by predetermined choices. In the back-

and-forth conversation of an interview, a researcher can ask for clarification, spend more time on a subtopic, or ask additional questions. In an interview, a subject will ideally feel free to open up and answer questions that are often complex. There are no right or wrong answers. The subject might not even know how to answer the questions honestly.

Questions such as, “How did society's view of alcohol consumption influence your decision whether or not to take your first sip of alcohol?” or “Did you feel that the divorce of your parents would put a social stigma on your family?” involve so many factors that the answers are difficult to categorize. A researcher needs to avoid steering or prompting the subject to respond in a specific way; otherwise, the results will prove to be unreliable. And, obviously, a sociological interview is not an interrogation. The researcher will benefit from gaining a subject’s trust, from empathizing or commiserating with a subject, and from listening without judgment.

Field Research

The work of sociology rarely happens in limited, confined spaces. Sociologists seldom study subjects in their own offices or laboratories. Rather, sociologists go out into the world. They meet subjects where they live, work, and play. **Field research** refers to gathering primary **data** from a natural environment without doing a lab experiment or a survey. It is a research method suited to an interpretive framework rather than to the scientific method. To conduct field research, the sociologist must be willing to step into new environments and observe, participate, or experience those worlds. In field work, the sociologists, rather than the subjects, are the ones out of their element.



Figure 1.5 Sociological researchers travel across countries and cultures to interact with and observe subjects in their natural environments. (Photo courtesy of IMLS Digital Collections and Content/flickr and Olympic National Park)

The researcher interacts with or observes a person or people and gathers data along the way. The key point in field research is that it takes place in the subject's natural environment, whether it's a coffee shop or tribal village, a homeless shelter or the DMV, a hospital, airport, mall, or beach resort.

While field research often begins in a specific *setting*, the study's purpose is to observe specific *behaviors* in that setting. Field work is optimal for observing *how* people behave. It is less useful, however, for understanding *why* they behave that way. You can't really narrow down cause and effect when there are so many variables floating around in a natural environment.

Much of the data gathered in field research are based not on cause and effect but on **correlation**. And while field research looks for correlation, its small sample size does not allow for establishing a causal relationship between two variables.

Sociology in the real world

Parrotheads as Sociological Subjects



Figure 1.6 Business suits for the day job are replaced by leis and T-shirts for a Jimmy Buffett concert. (Photo courtesy of Sam Howzitt/flickr)

Some sociologists study small groups of people who share an identity in one aspect of their lives. Almost everyone belongs to a group of like-minded people who share an interest or hobby. Scientologists, folk dancers, or members of Mensa (an organization for people with exceptionally high IQs) express a specific part of their identity through their affiliation with a group. Those groups are often of great interest to sociologists.

Jimmy Buffett, an American musician who built a career from his single top-10 song "Margaritaville," has a following of devoted groupies called Parrotheads. Some of them have taken fandom to the extreme, making Parrothead culture a lifestyle. In 2005, Parrotheads and their subculture caught the attention of researchers John Mihelich and John Papineau. The two saw the way Jimmy Buffett fans collectively created an artificial reality. They wanted to know how fan groups shape culture.

What Mihelich and Papineau found was that Parrotheads, for the most part, do not seek to challenge or even change society, as many sub-groups do. In fact, most Parrotheads live successfully within society, holding upper-level jobs in the corporate world. What they seek is escape from the stress of daily life.

At Jimmy Buffett concerts, Parrotheads engage in a form of role play. They paint their faces and dress for the tropics in grass skirts, Hawaiian leis, and Parrot hats. These fans don't generally play the part of Parrotheads outside of these concerts; you are not likely to see a lone Parrothead in a bank or library. In that sense, Parrothead culture is less about individualism and more about conformity. Being a Parrothead means sharing a specific identity. Parrotheads feel connected to each other: it's a group identity, not an individual one.

In their study, Mihelich and Papineau quote from a recent book by sociologist Richard Butsch, who writes, "un-self-conscious acts, if done by many people together, can produce change, even though the change may be unintended" (2000). Many Parrothead fan groups have performed good works in the name of Jimmy Buffett culture, donating to charities and volunteering their services.

However, the authors suggest that what really drives Parrothead culture is commercialism. Jimmy Buffett's popularity was dying out in the 1980s until being reinvigorated after he signed a sponsorship deal with a beer company. These days, his concert tours alone generate nearly \$30 million a year. Buffett made a lucrative career for himself by partnering with product companies and marketing Margaritaville in the form of T-shirts, restaurants, casinos, and an expansive line of products. Some fans accuse Buffett of selling out, while others admire his financial success. Buffett makes no secret of his commercial exploitations; from the stage, he's been known to tell his fans, "Just remember, I am spending your money foolishly."

Mihelich and Papineau gathered much of their information online. Referring to their study as a "Web ethnography," they collected extensive narrative material from fans who joined Parrothead clubs and posted their experiences on websites. "We do not claim to have conducted a complete ethnography of Parrothead fans, or even of the Parrothead Web activity," state the authors, "but we focused on particular aspects of Parrothead practice as revealed through Web research" (2005). Fan narratives gave them insight into how individuals identify with Buffett's world and how fans used popular music to cultivate personal and collective meaning.

In conducting studies about pockets of culture, most sociologists seek to discover a universal appeal. Mihelich and Papineau stated, "Although Parrotheads are a relative minority of the contemporary US population, an in-depth look at their practice and conditions illuminate [sic] cultural practices and conditions many of us experience and participate in" (2005).

Here, we will look at three types of field research: participant observation, ethnography, and the case study.

Participant Observation

In 2000, a comic writer named Rodney Rothman wanted an insider's view of white-collar work. He slipped into the sterile, high-rise offices of a New York "dot com" agency. Every day for two weeks, he pretended to work there. His main purpose was simply to see whether anyone would notice him or challenge his presence. No one did. The receptionist greeted him. The employees smiled and said good morning. Rothman was accepted as part of the team. He even went so far as to claim a desk, inform the receptionist of his whereabouts, and attend a meeting. He published an article about his experience in *The New Yorker* called "My Fake Job" (2000). Later, he was discredited for allegedly fabricating some details of the story and *The New Yorker* issued an apology. However, Rothman's entertaining article still offered fascinating descriptions of the inside workings of a "dot com" company and exemplified the lengths to which a sociologist will go to uncover material.

Rothman had conducted a form of study called **participant observation**, in which researchers join people and participate in a group's routine activities for the purpose of observing them within that context. This method lets researchers experience a specific aspect of social life. A researcher might go to great lengths to get a firsthand look into a trend, institution, or behavior. Researchers temporarily put themselves into roles and record their observations. A researcher might work as a waitress in a diner, live as a homeless person for several weeks, or ride along with police officers as they patrol their regular beat. Often, these researchers try to blend in seamlessly with the population they study, and they may not disclose their true identity or purpose if they feel it would compromise the results of their research.



Figure 1.7 Is she a working waitress or a sociologist conducting a study using participant observation? (Photo courtesy of zoetnet/flickr)

At the beginning of a field study, researchers might have a question: "What really goes on in the kitchen of the most popular diner on campus?" or "What is it like to be homeless?"

Participant observation is a useful method if the researcher wants to explore a certain environment from the inside.

Field researchers simply want to observe and learn. In such a setting, the researcher will be alert and open minded to whatever happens, recording all observations accurately. Soon, as patterns emerge, questions will become more specific, observations will lead to hypotheses, and hypotheses will guide the researcher in shaping data into results.

In a study of small towns in the United States conducted by sociological researchers John S. Lynd and Helen Merrell Lynd, the team altered their purpose as they gathered data. They initially planned to focus their study on the role of religion in U.S. towns. As they gathered observations, they realized that the effect of industrialization and urbanization was the more relevant topic of this social group. The Lynds did not change their methods, but they revised their purpose. This shaped the structure of *Middletown: A Study in Modern American Culture*, their published results (Lynd and Lynd 1959).

The Lynds were upfront about their mission. The townspeople of Muncie, Indiana, knew why the researchers were in their midst. But some sociologists prefer not to alert people to their presence. The main advantage of covert participant observation is that it allows the researcher access to authentic, natural behaviors of a group's members. The challenge, however, is gaining access to a setting without disrupting the pattern of others' behavior. Becoming an inside member of a group, organization, or subculture takes time and effort. Researchers must pretend to be something they are not. The process could involve role playing, making contacts, networking, or applying for a job.

Once inside a group, some researchers spend months or even years pretending to be one of the people they are observing. However, as observers, they cannot get too involved. They must keep their purpose in mind and apply the sociological perspective. That way, they illuminate social patterns that are often unrecognized. Because information gathered during participant observation is mostly qualitative, rather than quantitative, the end results are often descriptive or interpretive. The researcher might present findings in an article or book and describe what he or she witnessed and experienced.

This type of research is what journalist Barbara Ehrenreich conducted for her book *Nickel and Dimed*. One day over lunch with her editor, as the story goes, Ehrenreich mentioned an idea. *How can people exist on minimum-wage work? How do low-income workers get by?* she wondered. *Someone should do a study.* To her surprise, her editor responded, *Why don't you do it?*

That's how Ehrenreich found herself joining the ranks of the working class. For several months, she left her comfortable home and lived and worked among people who lacked, for the most part, higher education and marketable job skills. Undercover, she applied for and worked minimum wage jobs as a waitress, a cleaning woman, a nursing home aide, and a retail chain employee. During her participant observation, she used only her income from those jobs to pay for food, clothing, transportation, and shelter.

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/2-2-research-methods>

She discovered the obvious, that it's almost impossible to get by on minimum wage work. She also experienced and observed attitudes many middle and upper-class people never think about. She witnessed firsthand the treatment of working class employees. She saw the extreme measures people take to make ends meet and to survive. She described fellow employees who held two or three jobs, worked seven days a week, lived in cars, could not pay to treat chronic health conditions, got randomly fired, submitted to drug tests, and moved in and out of homeless shelters. She brought aspects of that life to light, describing difficult working conditions and the poor treatment that low-wage workers suffer.

Nickel and Dime: On (Not) Getting By in America, the book she wrote upon her return to her real life as a well-paid writer, has been widely read and used in many college classrooms.



Figure 1.8 Field research happens in real locations. What type of environment do work spaces foster? What would a sociologist discover after blending in? (Photo courtesy of drewzhrodague/flickr)

Ethnography

Ethnography is the extended observation of the social perspective and cultural values of an entire social setting. Ethnographies involve objective observation of an entire community.

The heart of an ethnographic study focuses on how subjects view their own social standing and how they understand themselves in relation to a community. An ethnographic study might observe, for example, a small U.S. fishing town, an Inuit community, a village in Thailand, a Buddhist monastery, a private boarding school, or an amusement park. These places all have borders. People live, work, study, or vacation within those borders. People are there for a certain reason and therefore behave in certain ways and respect certain cultural norms. An ethnographer would commit to spending a determined amount of time studying every aspect of the chosen place, taking in as much as possible.

A sociologist studying a tribe in the Amazon might watch the way villagers go about their daily lives and then write a paper about it. To observe a spiritual retreat center, an ethnographer might sign up for a retreat and attend as a guest for an extended stay, observe and record data, and collate the material into results.

Institutional Ethnography

Institutional ethnography is an extension of basic ethnographic research principles that focuses intentionally on everyday concrete social relationships. Developed by Canadian sociologist Dorothy E. Smith, institutional ethnography is often considered a feminist-inspired approach to social analysis and primarily considers women's experiences within male-dominated societies and power structures. Smith's work is seen to challenge sociology's exclusion of women, both academically and in the study of women's lives (Fenstermaker, n.d.).

Historically, social science research tended to objectify women and ignore their experiences except as viewed from the male perspective. Modern feminists note that describing women, and other marginalized groups, as subordinates helps those in authority maintain their own dominant positions (Social Sciences and Humanities Research Council of Canada, n.d.). Smith's three major works explored what she called "the conceptual practices of power" (1990; cited in Fenstermaker, n.d.) and are still considered seminal works in feminist theory and ethnography.

Sociological research

The Making of Middletown: A Study in Modern U.S. Culture

In 1924, a young married couple named Robert and Helen Lynd undertook an unprecedented ethnography: to apply sociological methods to the study of one U.S. city in order to discover what "ordinary" people in the United States did and believed. Choosing Muncie, Indiana (population about 30,000), as their subject, they moved to the small town and lived there for eighteen months.

Ethnographers had been examining other cultures for decades—groups considered minority or outsider—like gangs, immigrants, and the poor. But no one had studied the so-called average American.

Recording interviews and using surveys to gather data, the Lynds did not sugarcoat or idealize U.S. life (PBS). They objectively stated what they observed. Researching existing sources, they compared Muncie in 1890 to the Muncie they observed in 1924. Most Muncie adults, they found, had grown up on farms but now lived in homes inside the city. From that discovery, the Lynds focused their study on the impact of industrialization and urbanization.

They observed that Muncie was divided into business class and working class groups. They defined *business class* as dealing with abstract concepts and symbols, while *working class* people used tools to create concrete objects. The two classes led different lives with different goals and hopes. However, the Lynds observed, mass production offered both classes the same amenities. Like wealthy families, the working class was now able to own radios, cars, washing machines, telephones, vacuum cleaners, and refrigerators. This was an emerging material new reality of the 1920s.

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/2-2-research-methods>

As the Lynds worked, they divided their manuscript into six sections: Getting a Living, Making a Home, Training the Young, Using Leisure, Engaging in Religious Practices, and Engaging in Community Activities. Each chapter included subsections such as “The Long Arm of the Job” and “Why Do They Work So Hard?” in the “Getting a Living” chapter.

When the study was completed, the Lynds encountered a big problem. The Rockefeller Foundation, which had commissioned the book, claimed it was useless and refused to publish it. The Lynds asked if they could seek a publisher themselves.

Middletown: A Study in Modern American Culture was not only published in 1929 but also became an instant bestseller, a status unheard of for a sociological study. The book sold out six printings in its first year of publication, and has never gone out of print (PBS).

Nothing like it had ever been done before. *Middletown* was reviewed on the front page of the *New York Times*. Readers in the 1920s and 1930s identified with the citizens of Muncie, Indiana, but they were equally fascinated by the sociological methods and the use of scientific data to define ordinary people in the United States. The book was proof that social data was important—and interesting—to the U.S. public.



Figure 1.9 A classroom in Muncie, Indiana, in 1917, five years before John and Helen Lynd began researching this “typical” U.S. community. (Photo courtesy of Don O’Brien/flickr)

Case Study

Sometimes a researcher wants to study one specific person or event. A **case study** is an in-depth analysis of a single event, situation, or individual. To conduct a case study, a researcher examines existing sources like documents and archival records, conducts interviews, engages in direct observation and even participant observation, if possible.

Researchers might use this method to study a single case of, for example, a foster child, drug lord, cancer patient, criminal, or rape victim. However, a major criticism of the case study as a method is that a developed study of a single case, while offering depth on a topic, does not provide enough evidence to form a generalized conclusion. In other words, it is difficult to make universal claims based on just one person, since one person does not verify a pattern. This is why most sociologists do not use case studies as a primary research method.

However, case studies are useful when the single case is unique. In these instances, a single case study can add tremendous knowledge to a certain discipline. For example, a feral child, also called “wild child,” is one who grows up isolated from human beings. Feral children grow up without social contact and language, which are elements crucial to a “civilized” child’s development. These children mimic the behaviors and movements of animals, and often invent their own language. There are only about one hundred cases of “feral children” in the world.

As you may imagine, a feral child is a subject of great interest to researchers. Feral children provide unique information about child development because they have grown up outside of the parameters of “normal” child development. And since there are very few feral children, the case study is the most appropriate method for researchers to use in studying the subject.

At age three, a Ukrainian girl named Oxana Malaya suffered severe parental neglect. She lived in a shed with dogs, and she ate raw meat and scraps. Five years later, a neighbor called authorities and reported seeing a girl who ran on all fours, barking. Officials brought Oxana into society, where she was cared for and taught some human behaviors, but she never became fully socialized. She has been designated as unable to support herself and now lives in a mental institution (Grice 2011). Case studies like this offer a way for sociologists to collect data that may not be collectable by any other method.

Experiments

You’ve probably tested personal social theories. “If I study at night and review in the morning, I’ll improve my retention skills.” Or, “If I stop drinking soda, I’ll feel better.” Cause and effect. If this, then that. When you test the theory, your results either prove or disprove your hypothesis.

One way researchers test social theories is by conducting an **experiment**, meaning they investigate relationships to test a hypothesis—a scientific approach.

There are two main types of experiments: lab-based experiments and natural or field experiments. In a lab setting, the research can be controlled so that perhaps more data can be recorded in a certain amount of time. In a natural or field-based experiment, the generation of data cannot be controlled but the information might be considered more accurate since it was collected without interference or intervention by the researcher.

As a research method, either type of sociological experiment is useful for testing *if-then* statements: *if* a particular thing happens, *then* another particular thing will result. To set up a lab-based experiment, sociologists create artificial situations that allow them to manipulate variables.

Classically, the sociologist selects a set of people with similar characteristics, such as age, class, race, or education. Those people are divided into two groups. One is the experimental group and the other is the control group. The experimental group is exposed to the independent variable(s) and the control group is not. To test the benefits of tutoring, for

example, the sociologist might expose the experimental group of students to tutoring but not the control group. Then both groups would be tested for differences in performance to see if tutoring had an effect on the experimental group of students. As you can imagine, in a case like this, the researcher would not want to jeopardize the accomplishments of either group of students, so the setting would be somewhat artificial. The test would not be for a grade reflected on their permanent record, for example.

Sociological research

An Experiment in Action



Figure 1.10 Sociologist Frances Heussenstamm conducted an experiment to explore the correlation between traffic stops and race-based bumper stickers. This issue of racial profiling remains a hot-button topic today. (Photo courtesy of dwightsguest/flickr)

A real-life example will help illustrate the experiment process. In 1971, Frances Heussenstamm, a sociology professor at California State University at Los Angeles, had a theory about police prejudice. To test her theory she conducted an experiment. She chose fifteen students from three ethnic backgrounds: black, white, and Hispanic. She chose students who routinely drove to and from campus along Los Angeles freeway routes, and who'd had perfect driving records for longer than a year. Those were her independent variables—students, good driving records, same commute route.

Next, she placed a Black Panther bumper sticker on each car. That sticker, a representation of a social value, was the independent variable. In the 1970s, the Black Panthers were a revolutionary group actively fighting racism. Heussenstamm asked the students to follow their normal driving patterns. She wanted to see whether seeming support of the Black Panthers would change how these good drivers were treated by the police patrolling the highways. The dependent variable would be the number of traffic stops/citations.

The first arrest, for an incorrect lane change, was made two hours after the experiment began. One participant was pulled over three times in three days. He quit the study. After seventeen days, the fifteen drivers had collected a total of thirty-three traffic citations. The

experiment was halted. The funding to pay traffic fines had run out, and so had the enthusiasm of the participants (Heussenstamm 1971).

Secondary Data Analysis

While sociologists often engage in original research studies, they also contribute knowledge to the discipline through **secondary data analysis**. Secondary data doesn't result from firsthand research collected from primary sources, but are the already completed work of other researchers. Sociologists might study works written by historians, economists, teachers, or early sociologists. They might search through periodicals, newspapers, or magazines from any period in history.

Using available information not only saves time and money but can also add depth to a study. Sociologists often interpret findings in a new way, a way that was not part of an author's original purpose or intention. To study how women were encouraged to act and behave in the 1960s, for example, a researcher might watch movies, television shows, and situation comedies from that period. Or to research changes in behavior and attitudes due to the emergence of television in the late 1950s and early 1960s, a sociologist would rely on new interpretations of secondary data. Decades from now, researchers will most likely conduct similar studies on the advent of mobile phones, the Internet, or Facebook.

Social scientists also learn by analyzing the research of a variety of agencies. Governmental departments and global groups, like the U.S. Bureau of Labor Statistics or the World Health Organization, publish studies with findings that are useful to sociologists. A public statistic like the foreclosure rate might be useful for studying the effects of the 2008 recession; a racial demographic profile might be compared with data on education funding to examine the resources accessible by different groups.

One of the advantages of secondary data is that it is **nonreactive research** (or unobtrusive research), meaning that it does not include direct contact with subjects and will not alter or influence people's behaviors. Unlike studies requiring direct contact with people, using previously published data doesn't require entering a population and the investment and risks inherent in that research process.

Using available data does have its challenges. Public records are not always easy to access. A researcher will need to do some legwork to track them down and gain access to records. To guide the search through a vast library of materials and avoid wasting time reading unrelated sources, sociologists employ **content analysis**, applying a systematic approach to record and value information gleaned from secondary data as they relate to the study at hand.

But, in some cases, there is no way to verify the accuracy of existing data. It is easy to count how many drunk drivers, for example, are pulled over by the police. But how many are not? While it's possible to discover the percentage of teenage students who drop out of high school, it might be more challenging to determine the number who return to school or get their GED later.

Another problem arises when data are unavailable in the exact form needed or do not include the precise angle the researcher seeks. For example, the average salaries paid to professors at a public school is public record. But the separate figures don't necessarily reveal how long it took each professor to reach the salary range, what their educational backgrounds are, or how long they've been teaching.

When conducting content analysis, it is important to consider the date of publication of an existing source and to take into account attitudes and common cultural ideals that may have influenced the research. For example, Robert S. Lynd and Helen Merrell Lynd gathered research for their book *Middletown: A Study in Modern American Culture* in the 1920s. Attitudes and cultural norms were vastly different then than they are now. Beliefs about gender roles, race, education, and work have changed significantly since then. At the time, the study's purpose was to reveal the truth about small U.S. communities. Today, it is an illustration of 1920s' attitudes and values.

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/2-2-research-methods>

1.3 Ethical concerns

Sociologists conduct studies to shed light on human behaviors. Knowledge is a powerful tool that can be used toward positive change. And while a sociologist's goal is often simply to uncover knowledge rather than to spur action, many people use sociological studies to help improve people's lives. In that sense, conducting a sociological study comes with a tremendous amount of responsibility. Like any researchers, sociologists must consider their ethical obligation to avoid harming subjects or groups while conducting their research.

The American Sociological Association, or ASA, is the major professional organization of sociologists in North America. The ASA is a great resource for students of sociology as well. The ASA maintains a **code of ethics**—formal guidelines for conducting sociological research—consisting of principles and ethical standards to be used in the discipline. It also describes procedures for filing, investigating, and resolving complaints of unethical conduct.

Practicing sociologists and sociology students have a lot to consider. Some of the guidelines state that researchers must try to be skillful and fair-minded in their work, especially as it relates to their human subjects. Researchers must obtain participants' informed consent and inform subjects of the responsibilities and risks of research before they agree to partake. During a study, sociologists must ensure the safety of participants and immediately stop work if a subject becomes potentially endangered on any level.

Researchers are required to protect the privacy of research participants whenever possible. Even if pressured by authorities, such as police or courts, researchers are not ethically allowed to release confidential information. Researchers must make results available to other sociologists, must make public all sources of financial support, and must not accept funding from any organization that might cause a conflict of interest or seek to influence the research results for its own purposes. The ASA's ethical considerations shape not only the study but also the publication of results.

Pioneer German sociologist Max Weber (1864–1920) identified another crucial ethical concern. Weber understood that personal values could distort the framework for disclosing study results. While he accepted that some aspects of research design might be influenced by personal values, he declared it was entirely inappropriate to allow personal values to shape the interpretation of the responses. Sociologists, he stated, must establish **value neutrality**, a practice of remaining impartial, without bias or judgment, during the course of a study and in publishing results (1949). Sociologists are obligated to disclose research findings without omitting or distorting significant data.

Is value neutrality possible? Many sociologists believe it is impossible to set aside personal values and retain complete objectivity. They caution readers, rather, to understand that sociological studies may, by necessity, contain a certain amount of value bias. It does not discredit the results but allows readers to view them as one form of truth rather than a singular fact. Some sociologists attempt to remain uncritical and as objective as possible when studying cultural institutions. Value neutrality does not mean having no opinions. It means striving to overcome personal biases, particularly subconscious biases, when analyzing data. It means avoiding skewing data in order to match a predetermined outcome that aligns with a particular

agenda, such as a political or moral point of view. Investigators are ethically obligated to report results, even when they contradict personal views, predicted outcomes, or widely accepted beliefs.

1.4 Summary of research methods

Method	Implementation	Advantages	Challenges
Survey	• Questionnaires • Interviews	• Yields many responses • Can survey a large sample • Quantitative data are easy to chart	• Can be time consuming • Can be difficult to encourage participant response • Captures what people think and believe but not necessarily how they behave in real life
Field Work	• Observation • Participant observation • Ethnography • Case study	• Yields detailed, accurate real-life information	• Time consuming • Data captures how people behave but not what they think and believe • Qualitative data is difficult to organize
Experiment	• Deliberate manipulation of social customs and mores	• Tests cause and effect relationships	• Hawthorne Effect • Ethical concerns about people's wellbeing
Secondary Data Analysis	• Analysis of government data (census, health, crime statistics) • Research historic documents	• Makes good use of previous sociological information	• Data could be focused on a purpose other than yours • Data can be hard to find

Table2.2 Main Sociological Research Methods Sociological research methods have advantages and disadvantages.

References Chapter 1

American Sociological Association. (1999). *Code of Ethics*. Retrieved on July 1, 2011 from <http://www.asanet.org/about/ethics.cfm>.

Arkowitz, H. & Lilienfeld, S.O. (2009). Lunacy and the full moon: does a full moon really trigger strange behavior? *Scientific American*. Retrieved on December 30, 2014 <http://www.scientificamerican.com/article/lunacy-and-the-full-moon/>.

Arkowitz, H. & Lilienfeld, S.O. (2009). Lunacy and the full moon: does a full moon really trigger strange behavior? *Scientific American*. Retrieved on October 20, 2014 <http://www.scientificamerican.com/article/lunacy-and-the-full-moon>.

Berger, P.L. (1963). *Invitation to Sociology: A Humanistic Perspective*. New York: Anchor Books.

Butsch, R. (2000). *The Making of American Audiences: From Stage to Television, 1750–1990*. Cambridge: Cambridge UP.

Caplow, T., Hicks, L. & Wattenberg, B. (2000). The First Measured Century: Middletown. *The First Measured Century*. PBS. Retrieved on February 23, 2012 from <http://www.pbs.org/fmc/index.htm>.

Durkheim, É. (1897). *Suicide: A Study in Sociology*. Reprint, New York: Free Press, 1966.

Fenstermaker, S. n.d. Dorothy E. Smith Award Statement. *American Sociological Association*. Retrieved on October 19, 2014 from <http://www.asanet.org/about/awards/duboiscaer/smith.cfm>.

Franke, R.H. & Kaul, J.D. (1978). The Hawthorne Experiments: First Statistical Interpretation. *American Sociological Review*, 43(5), 632–643.

Grice, E. (2006). Cry of an enfant sauvage. *The Telegraph*. Retrieved on July 20, 2011 from <http://www.telegraph.co.uk/culture/tvandradio/3653890/Cry-of-an-enfant-sauvage.html>.

Heussenstamm, F.K. (1971). Bumper stickers and cops. *Trans-action: Social Science and Modern Society*, 4, 32–33.

Igo, S.E. (2008). *The averaged American: surveys, citizens, and the making of a mass public*. Cambridge, MA: Harvard University Press.

Lynd, R.S. & Lynd, H.M. (1959). *Middletown: A Study in Modern American Culture*. San Diego, CA: Harcourt Brace Javanovich.

This chapter is reproduced with small alterations from taken from **Introduction to Sociology** (2nd edition), written by Heather Griffiths and Nathan Keirns (Houston, TX: OpenStax, 2020), licensed under CC BY 4.0. Access for free at <https://openstax.org/books/introduction-sociology-2e/pages/2-references>

- Lynd, S. (2005). Making Middleton. *Indiana Magazine of History*, 101(3), 226–238.
- Merton, R. (1949). *Social Theory and Social Structure*. Reprint, New York: Free Press, 1968.
- Mihelich, J. & Papineau, J. (2005). Parrotheads in Margaritaville: fan practice, oppositional culture, and embedded cultural resistance in Buffett fandom. *Journal of Popular Music Studies*, 17(2), 175–202.
- Pew Research Center. (2014). Ebola worries rise, but most are 'fairly' confident in government, hospitals to deal with disease: broad support for U.S. efforts to deal with Ebola in West Africa. *Pew Research Center for the People & the Press*, Retrieved on October 25, 2014 from <https://www.pewresearch.org/politics/2014/10/21/ebola-worries-rise-but-most-are-fairly-confident-in-government-hospitals-to-deal-with-disease/>.
- Rossi, P.H. (1987). No good applied social research goes unpunished. *Society*, 25(1), 73–79.
- Rothman, R. (2000). My fake job. *The New Yorker*, November 27, 2000.
- Rotton, J. & Kelly, I.W. (1985). Much ado about the full moon: a meta-analysis of lunar-lunacy research. *Psychological Bulletin*, 97(2), 286–306.
- Social Sciences and Humanities Research Council of Canada. n.d. Institutional Ethnography. Retrieved on October 19, 2014 from <http://web.uvic.ca/~mariecam/kgSite/institutionalEthnography.html>.
- Sonnenfeld, J.A. (1985). Shedding light on the Hawthorne studies. *Journal of Occupational Behavior*, 6, 125.
- University of Utah. (2012). *Scientific Method Lab*. Retrieved from http://aspire.cosmic-ray.org/labs/scientific_method/sci_method_main.html.
- Weber, M. (1949). *Methodology of the Social Sciences*. Translated by H. Shils and E. Finch. Glencoe, IL: Free Press.

Chapter 2: What are spatial sciences?

Geography is a discipline capable of helping those who know how to use it to better understand a variety of subjects and topics. Geography is a set of tools that help you ask questions, see patterns in data, solve problems and communicate solutions.



Figure 2.1 Far too many people equate geography with trivia, undermining a real understanding of what geography is as a discipline. (Photo courtesy of SethAllen623/Wikimedia)

The popular afternoon television show Jeopardy is probably the most common way Americans are exposed to geography. This is a huge problem for the discipline of geography because although Jeopardy does more than any other medium to advance geographic knowledge among Americans, it advances our understanding of Geography down a dead-end street. A typical geography question on Jeopardy might ask contestants to identify the capital of Nebraska or a mountain range in Switzerland. Professional geographers rarely ask questions of that sort. Jeopardy, because it constantly puts a spotlight on “*geography as subject*”, effectively reinforces old-fashioned notions about geography as trivia, and in the process leads many Americans to believe that geographers mainly memorize maps, weather patterns, and capital cities. Of course, some amount of memorization of facts is necessary, but to think it’s the focus of Geography is like thinking Historians only memorize dates, and English majors spend all their time preparing for spelling bees.

For generations, in America, K-12 curricula and the accompanying textbooks have echoed the unfortunate focus upon geography-as-trivia promoted by TV game shows. As a result, the much more accurate notion of “*geography as a discipline*” has been all but eliminated from the American imagination.

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at

<https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>

Subsequently, college students rarely consider geography as a major. Many, including high school guidance counselors, do not even realize that one can major in geography at most universities. Students, parents and even faculty outside of your Geography Department often are unaware that geography provides students and scholars with a robust set of analytical tools that can be applied to many topics of interest, and lead to lucrative career paths in a stunning array of private industries, non-profits and with many public agencies.

One of the primary goals of this text is to introduce readers to an updated, viable version of geography. Hopefully, students will come to think of geography primarily as a **discipline** that permits those who have mastered it to study virtually any topic that takes place. By the end of the semester, students using this text should begin to be able to: 1) see the world like a geographer, 2) ask questions like a geographer, 3) solve problems like a geographer and 4) communicate like a geographer....and maybe you'll learn some old-school geography trivia to defeat friends at Jeopardy!

2.2 A Short History of Geography

Geography is sometimes called the “mother of all disciplines” because it has been around for so long. Although the discipline has undergone major changes in the last few decades, most people still have very old-fashioned ideas about geography. Those old-fashioned ideas still dominate much K-12 instruction in Geography and are partly a result of the tumultuous history of Geography.

Ancient Greek and Chinese scholars wrote massive “geographies” that defined the discipline for centuries. Until the 1800s, most “geographers” spent much of their time writing highly descriptive or **idiographic** narratives about various regions of the world. Descriptive geographies are interesting for those who have a healthy intellectual curiosity about the people, places, and cultures of the world, and descriptive geographies proved immeasurably valuable to imperialists, colonizers, and military planners. Unfortunately, in the context of schooling, descriptive geographies often tend to degenerate into an encyclopedic list of facts about a region or location.

Environmental Determinism

As the methodologies of science, and indeed social science, evolved during the 19th century, the production of mere descriptions of regions and locations fell short of what geographers (and others) thought worthy of true scholarship. Some geographers tried to make the discipline more scientific by seeking to identify and describe **causal** connections between group behaviors and local environmental conditions. In other words, they sought to understand the cause-and-effect relationship between culture and the physical environment. This type of geography became known as environmental determinism. **Environmental determinists** sought to demonstrate how local conditions like climate, topography and soil characteristics were key **determinants**, or causes, behind the evolution of local or regional cultural practices. Perhaps not surprisingly, many environmental determinist scholars found that their culture group (i.e. Europeans) were products of ideal environmental conditions.

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at <https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>

Environmental determinists argued that the most advanced societies developed where favorable environmental conditions existed, i.e. North America, Europe. According to their logic, locations that were too hot, too cold, too rainy, too dry, etc., produced inferior societies and inferior people. For example, Ellen Churchill Semple wrote in 1911, "among mountain as among desert peoples, robbery tends to become a virtue; environment dictates their ethical code". The bigotry implicit and explicit in environmental determinism is clearer today than it was 100 years ago. Despite an inability to prove scientifically their theories, a few geographers, like Semple and Ellsworth Huntington commanded large audiences in the early 20th century.

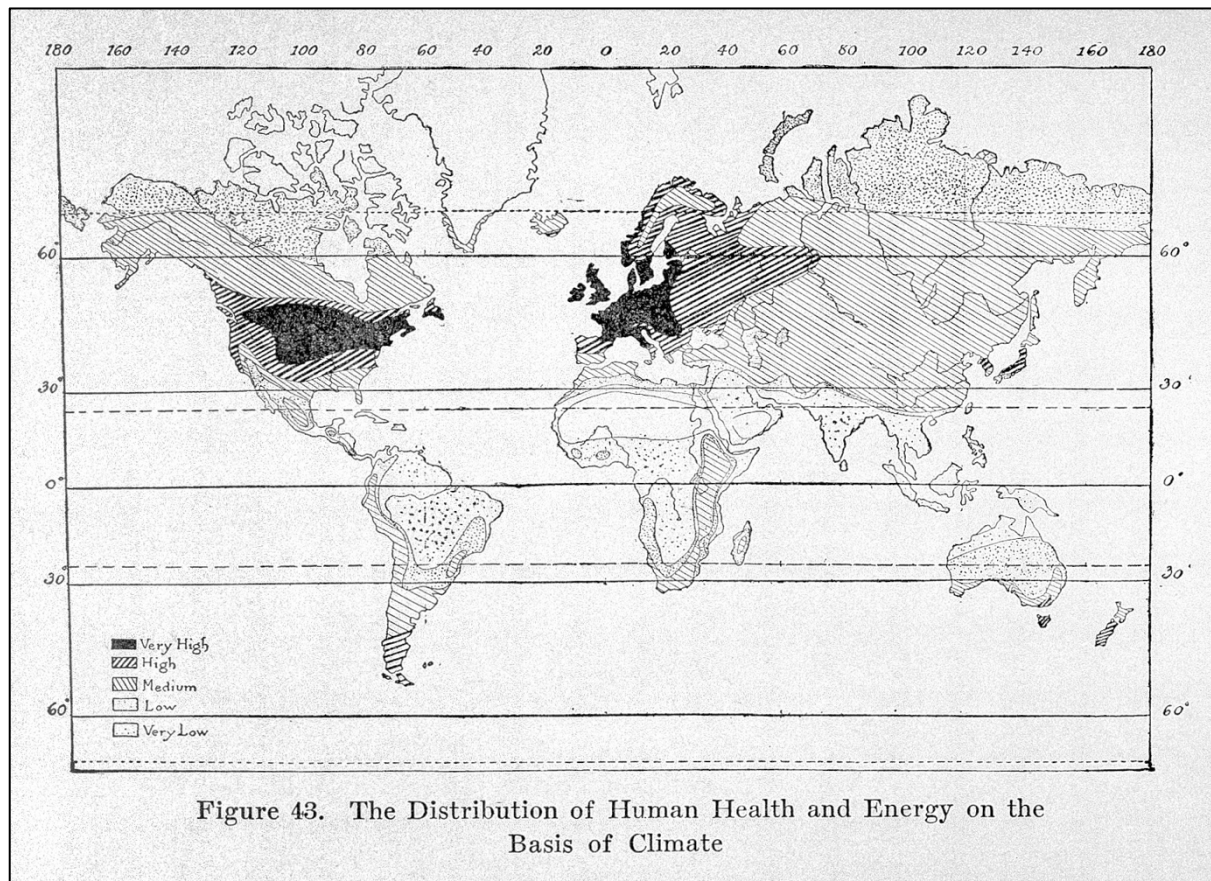


Figure 2.2 Map – An environmental determinist’s map of “health and human energy” based on climatic predictors. Very high scores were assigned to mid-latitude climates (Europe, US), low scores to tropical and equatorial climates. (Photo courtesy of Wellcome Images/Wikimedia)

Although many academic geographers forcefully rejected the ideas and concepts advanced by environmental determinists, it wasn’t really until World War II that a solid majority of people understood the horrifying implications that this brand of pseudo-scientific scholarship had upon the *eugenics* movement and the evolution of Fascism. Geography’s deep involvement with Environmental Determinism tainted the reputation of the discipline for decades.

Paradigm Wars – 1950-1990

The reaction to Environmental Determinism by a significant number of geographers was to turn away from scientific methods so ill-used by the Environmental Determinists. Some called for a return to the descriptive or ideographic geography of the past. Today, this type of geography is largely represented by an approach known as ***regional geography***, and it remains the dominant version of geography taught at the K-12 level. Done well, regional geography can be an exciting and intellectually stimulating exercise capable of providing a pathway to understanding the unique dynamics within regions, why various locations on earth differ from other places, and why many places have similarities. Unfortunately, regional geography in classroom settings often degrades into a “forced march” of endless memorization of facts about far-off locations. Most K-12 school curricula and many introductory college courses fall into this trap. Students in regional style courses often learn little about why places have become unique because poorly designed curricula fail to focus student attention on the processes driving the creation or evolution of regions. The dominance of the regional approach has generated the false, but the popular perception of geography as trivia, especially in America. In Europe, geography did not fall into that trap.

By the 1960s, Geography Departments were facing elimination from many college campuses. As a result, many geographers began adopting legitimate scientific methodologies via spatial statistics during a period known today as the ***quantitative revolution***, a major ***paradigm shift*** that continues to accelerate today, especially among users of Geographic Information Systems (GIS –see below). Today, most geographers seek to discover generalizable patterns, or even laws, that describe or govern society. This is the ***nomothetic*** approach. Coupled with accompanying revolutions in our ability to collect, store, manipulate and analyze spatial data, geographers today are engaged in complex, high-tech research on a wide array of pressing issues often using so-called “big data”.

Beginning in the 1970s and accelerating rapidly in the 1990s, geographers also occupied leading roles in a thrilling expansion of spatially informed theoretical approaches attempting to explain how the world works. Important, economic, political and cultural theorists emerged from among the ranks of geography departments in the UK and the US, playing important roles in an overall flowering of ***critical geography***, during a period known as the cultural turn within geography. Many geographers today focus squarely on the complex, often subtle, mechanics that regulate the production and maintenance of knowledge itself, which is in some ways the final frontier of social science. Critical geographers work to uncover how and why societies and individuals believe what they do, and how those beliefs are manufactured, manipulated, distorted, maintained, subverted, appropriated or eliminated.

Today, geography is a very vibrant discipline offering to the uninitiated, a surprising number of avenues to understand the world, as well as multiple pathways to high-paying jobs in the public and private sectors. Geographers make a difference in the world.

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at <https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>

2.2 Geography as Discipline – Key Aspects of the “Jedi Way”

If you go to the library at your college or university and head to the section housing books about geography, you may be disappointed to find there’s almost nothing there. You might mistakenly believe that geographers don’t write books, or that geography is exceptionally limited in its scope. Both assumptions would be wrong. Libraries have lots of books written by geographers, but because geographers can study almost any subject, books written by geographers are found scattered throughout the library, because libraries are organized by the subject of study, not the method by which subjects are studied and analyzed.

Geographers write books and publish scholarly articles about an astounding array of subjects – far too many to consider here. The main point is for you to remember is that geographers can study almost anything that takes place. What are you interested in? If it exists someplace, you can bet some geographer has studied it. The author of this text wrote a dissertation on innovation waves in pop music (rock, rap, country), but has since researched subprime lending, crime, health problems, gentrification, homelessness and other topics of interest.

The section that follows provides a unique, but basic, guide to help you better understand geography as a discipline. The entire text is designed around a set of guiding principles that characterize the way geographers tend to observe, query and analyze the world around them – and how we communicate findings and theories. To better help you learn to observe, question and analyze the world around you, this text leverages a variety of similes and metaphors from Star Wars movies. The text may refer to geographers as “Jedi” because geographers, like the Jedi in the Star Wars movies, have a special and powerful way of doing things (observing, questioning, analyzing, and communicating) that, at least in the academic and professional worlds, gives geographers unique abilities to gain insights often overlooked by others.

Geography is a Way to See the World: Jedi Goggles

The ability to “read” the landscape is the first skill of the Jedi-Geographer. It sounds simple enough, but advanced ability in ***landscape interpretation*** takes many years to develop. Generally, people are not aware that all landscapes, including both the human-built and the natural, physical environment, can be read, somewhat like you read text in a book. With some practice, you can develop a measure of ***landscape literacy***, which is the ability to “read” and understand a significant number of messages inscribed into the environment. All landscapes tell a story. Throughout this text, there are opportunities for you to learn to read them. This skill is referred to as using your “Jedi Goggles.” Noted scholar Aldo Leopold also thought this an important skill, noting, “The objective is to teach the student(s) to see the land, to understand what [they see], and enjoy what [they] understand...”

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at

<https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>



Figure 2.3 Las Vegas, Nevada. Casino landscapes often evoke exotic locations to heighten the excitement of gamblers, who then are more likely to spend more prolifically once the landscape conditions their mood properly. Map – An environmental determinist’s map of “health and human energy” based on climatic predictors. Very high scores were assigned to mid-latitude climates (Europe, US), low scores to tropical and equatorial climates. ([Photo courtesy of Tourist Landscapes/Google Photos](#))

Certainly, you already have some skill reading the landscape, but most non-geographers do not read the landscape well because they do not think to do it consciously or regularly. Like a Jedi without training, you have occasionally put on “Jedi Goggles” but maybe didn’t think about what you were doing. For example, you have probably at some point found yourself driving in a dangerous neighborhood. Of course, there are no road signs at the borders of a high crime neighborhood warning, “Caution – Now Entering Dangerous Neighborhood”. Still, you know to be cautious. Why? Because- you have developed some skill reading the landscape over the years. Rubbish on the street, graffiti, bars on windows and unkempt lawns are common landscape markers (visual cues) of a neighborhood that suffers from crime; your sense of self-preservation has prompted you to learn to interpret landscape symbols associated with danger. Many police and people who are “street smart” have well-developed “Jedi Goggles”.

The first step you must take as you develop landscape literacy skills is to begin actively looking more closely at stores, rivers, houses, parking lots, road signs, empty lots, farm fields, hillsides and anything else that passes your windshield. As you look, think to yourself, “What is this landscape telling me?” There’s no need to focus on “pretty” landscapes because beautiful, scenic, impressive or extraordinary landscapes are no more important than those that are boring, ugly or commonplace. The ordinary, everyday landscapes, called **vernacular landscapes** are, as Peirce Lewis noted, our “unwitting autobiography, reflecting our tastes, our values, our aspirations and even our fears in tangible visible form.” There is a reassuring honesty one gets reading vernacular landscapes. The lack of intentionality makes landscapes a source of unvarnished truth about a place’s history, unlike purposefully written histories that are often prone to biases and outright lies.

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at

<https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>

This text will help you learn to read the cultural histories of various places and regions as they have been written on the landscape. Hopefully, by the end of the semester, you'll find yourself asking "What does this scene tell me about this place?" Hopefully, you'll begin to see patterns you never noticed before and you'll be trying to guess what forces created that which you see. With luck, you may find yourself asking "Why is this here, and not elsewhere?" Each chapter in this book features multiple images of landscapes. Most images are captioned with a quick "reading" of the images. A large photographic database, available online at The American Landscape Project, has many thousands of the images used in this text and many feature robust captions to help you learn to confidently read landscapes.

Geography is a Way to Ask Questions – Jedi MindTricks

Geographers also have a unique way of knowing what we know. **Epistemology** is the study of knowledge and because geographers have a unique way of thinking about questions and arriving at answers, we have a unique epistemology. Most of the time, people don't think about how they know what they know. We don't think about our epistemological processes, biases, and tendencies. Therefore, we don't often know how we know, what we know. "I just know it!" is how you might respond if someone challenged you on your epistemology. For scientists, social scientists and other serious thinkers, "I just know it" is an unacceptable response. It's important to understand one's epistemological tendencies. *You must know how you know, what you know.*



Figure 2.4 Painting – The story of the blind men and the elephant illustrates how knowledge is dependent upon the perspective of the individual and the imperfect ability of anyone to understand the entire truth of any subject or circumstance. Las Vegas, Nevada. Casino landscapes often evoke exotic locations to heighten the excitement of gamblers, who then are more likely to spend more prolifically once the landscape conditions their mood properly. Map – An environmental determinist's map of "health and human energy" based on climatic predictors. Very high scores were assigned to mid-latitude climates (Europe, US), low scores to tropical and equatorial climates. ([Hanabusa Itchō/Library of Congress/Wikimedia](https://www.loc.gov/rr/humanities/itcho/itcho.html))

Geographers have a favored epistemology – it doesn't have a name, but it does begin with a simple strategy that leads us to learn and/or understand things differently than non-geographers. When geographers seek to understand the world around them better, we have

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at <https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>

a strong tendency to frame questions and answers spatially. In other words, when geographers want to know “Why?” or “How?”, we first typically ask the question “Where?”. Privileging spatial questions in a quest to understand the world is like “Using the Force” among Jedi warriors. When geographers “Use the Force,” they ask “Where?” to understand “Why”. By doing so, geographers often come to different conclusions than non-geographers. By asking “Where?” when you want to know “Why” or “How” is the most basic of all “Jedi Mind Tricks”. Historians have a similar trick - they ask “When?” (rather than “Where?”) while seeking understanding. Still, learning to ask questions about how the world works consistently is crucial to developing a “habit of mind”; itself a crucial element in developing a disciplinary epistemology – something all college students should work to develop. Faculty in many majors do not teach students to develop disciplinary habits-of-mind. The main goal of this course is to help you develop this habit-of-mind.

Most people don’t use “The Force” and there are consequences for understanding our world. Consider for instance a conversation overheard by the author of this text some years ago on a large college campus in Illinois. I overheard two people discussing why most African-American students on that campus rode the elevator in a campus building while most white students used stairs instead. It was evident and obvious that this was occurring, but the causal reasons for this difference in behavior eluded those discussing the behaviors. One person in the conversation was quick to assign ethnicity as the primary causal factor motivating students to use or avoid elevators. A geographer using “The Force” would not have made this mistake because the geographer would have thought about the spatial aspects of the phenomenon first, and would have realized that ethnicity was probably not an important factor explaining why black and white students used elevators differently. Instead, a geographer would have asked, “**where** are the students who use elevators from?” Had the people discussing the question asked “Where?”, it may have occurred to them that many of the black students on that campus grew up in Chicago, where using elevators to reach the upper floors of tall residential apartment buildings was a daily ritual. The behavioral habits developed by students living in high-rise apartments lingered with students after they moved to campus, prompting them to select the elevator to navigate to upper stories on campus out of habit, even when the building was only two or three stories. Students from rural and suburban locations, on the other hand, typically lived in single-story or two-story houses, and were, therefore, more accustomed to reaching an upper floor by climbing stairs out of force of habit. To a geographer, those behavioral differences were not black and white – but spatial. Thinking that way, at least by Steve Graves, is called “Using the Force”.

By privileging matters of place and space in their quest for knowledge, geographers may bias their conclusions – arguing that location, space or place is a significant causal variable in the outcome of various phenomena. Favoring one epistemology over others is nearly unavoidable, and it is acceptable if one recognizes the biases an epistemology creates. Historians, sociologists, political scientists, economists, etc. each have their own epistemological biases. If these competing means to comprehend reality can be given a fair audience, a robust, multi-perspectival or **interdisciplinary** understanding of our world is possible. There is danger in discounting epistemologies unfamiliar to you – and because the US school system does a poor job of developing students’ spatial epistemology, taking a

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at <https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>

geography course and developing some measure of this unique habit of mind is a critical element in a quality, robust liberal arts education.

Geography is a Way to Solve Problems – Jedi Weapons

Geographers have a very powerful toolbox of problem-solving tactics and strategies. The tools in the geographer's toolbox are our methods and most of our methods are dependent upon our spatial epistemology, or way of knowing. There are dozens of methods used by geographers today. This is partly because geographers have adopted many methods used by scholars in the sciences, social sciences, and the humanities. The **scientific method** is widely used by geographers, but humanistic methods, like those employed by historians or even art critics, are also used by geographers. Generally, geographers alter these borrowed methods so we can use them effectively alongside our discipline's spatial epistemology. Taken together, the methods and the epistemology provide geographers with a series of **methodologies**; or set of rules that govern the collection and analysis of a wide variety of data.

For example, if a geographer were to survey students about a new campus policy, many of the questions included in the survey would be the same as those used by a political scientist, historian or sociologist. However, a geographer would insist that the survey include a spatial question among the demographic questions. Instead of simply asking about each survey respondent's age, gender, and ethnicity; the geographer would likely insist on asking about each respondent's address, ZIP code, or at least "hometown".

Like many other disciplines, geographers use **statistics**. Often, we use statistics in a manner indistinguishable from the way they are used in other disciplines. Other times we have found a need to develop separate geography-friendly statistical tools more suited to answering spatial questions. For example, social scientists from other disciplines might start a statistical inquiry by calculating the mean, median and standard deviation of some data points. Geographers, on the other hand, might first plot the data on a map, and then calculate a spatial mean, spatial median and the standard distance of the same data. That's because geographers want to know where and why. There are a vast number of spatial statistics. Some are exceptionally complex and some quite simple. **Spatial Analysis** is the name of a subfield of geography which applies advanced mathematical analyses to spatial data. There are a lot of good jobs for people who can do spatial analyses of data.

For the past few decades, the primary tool in the geographer's toolbox has been a suite of software products known as **Geographic Information Systems** or **GIS**. GIS is the "lightsaber" of the modern geographer. Like the lightsaber used by Jedi in Star Wars movies, GIS is used by a select few (Jedis), and in the hands of a "master", GIS is both amazingly powerful and versatile. GIS software allows geographers to analyze data in a unique fashion, permitting geographers to ask spatial questions and solve problems using our special methods and while engaging our peculiar spatial epistemology. GIS allows geographers to sometimes solve problems that have proven intractable to those without GIS. People from other fields begun to embrace GIS, but the techniques for using GIS have begun to emerge as foundational tools for modern geographers.

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at <https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>

Geography is a Way to Communicate – Cartography the Jedi Language

Because you are reading this, you have some command of the rules that govern the English language. You no doubt can write in this language as well, another indication of your textual literacy: i.e. you can communicate with a written language. You probably are reasonably adept at communicating with numbers as well – so you are numerate or you have numeracy skills. Without these two key skillsets, you would not be in college. Geographers must be literate and numerate, but we also work to heighten our ability to communicate with non-textual, visual imagery, and/or graphics as well. Geographers who can both “read” graphics and create “readable” graphics have graphicacy skills and are considered graphicate. Persons who can read and create legible, communicative maps, are considered cartographicate and possess cartographicacy skills. Mapmakers are called cartographers.

Cartography is the science and art of map-making and it is the specialized language of the geographer. The ability to communicate vast amounts of information, and/or ideas using maps is an excellent skill to develop. In the last decade, thanks to Google Maps/Earth, GPS and a massive increase in the value placed on spatial data by the government, military and private enterprise, cartographicacy has blossomed as a valuable type of literacy.

It makes sense that maps have re-entered the public’s consciousness in recent years. Besides the explosion in spatial data available with which cartographers can create fun or informative maps, changes in the pace of our lives as well the tendency for us to be overwhelmed by data in the digital age have made well-constructed maps and graphics a welcome coping mechanism. Maps allow our brains to rapidly process far greater amounts of data than we could if we encountered the same data as text or in a massive spreadsheet. Maps allow us to see patterns and processes that would be difficult to discern otherwise.

Although some folks find maps innately easy to read, maps confuse others. Moreover, it can be very challenging to author a legible map. Good cartography is harder than it looks. Some colleges feature full programs in Cartography. Professional cartography is a great career path for the talented few who master this Jedi art. All geographers, even those whose specialty is not cartography, should endeavor to become reasonably adept at making maps. Thanks to GIS, most people who know GIS can create a tolerably good map using the software’s built-in cartography templates.

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at <https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>

References chapter 2

Fraser, H.J. (1982). The highest form of the geographer's art. *Annals of the Association of American Geographers*, 72(1), 1-29.

Graves, S.M. (2013). Geography as a 'Jedi's discipline'. *The California Geographer*, 53, xx-xx.

Huntington, E. (1915). *Civilization and Climate*. New Haven: Yale University Press.

Lewis, P.F. (1979) Axioms for reading the landscape. *The interpretation of ordinary landscapes*, 23, 167-187

Patterson, T. (2010). Outside the bubble: Real-world mapping advice for students. *Cartographic Perspectives*, 65, 7-15.

Semples, E.C. (1911) *Influences of geographic environment: On the basis of Ratzel's System of Anthro-Geo-Geography*.

This chapter is reproduced with small alterations from **Introduction to Human Geography: A Disciplinary Approach** (3rd edition), written by Steven Graves. Access for free at <https://drive.google.com/file/d/1PCzXn7zgOEAA2eQOvuCAibWkTLjQOdGk/view>

Chapter 3: Theories in scientific research

As we know from previous chapters, science is knowledge represented as a collection of “theories” derived using the scientific method. In this chapter, we will examine what is a theory, why do we need theories in research, what are the building blocks of a theory, how to evaluate theories, how can we apply theories in research, and also presents illustrative examples of five theories frequently used in social science research.

Theories

Theories are explanations of a natural or social behavior, event, or phenomenon. More formally, a scientific theory is a system of constructs (concepts) and propositions (relationships between those constructs) that collectively presents a logical, systematic, and coherent explanation of a phenomenon of interest within some assumptions and boundary conditions (Bacharach 1989).

Theories should explain why things happen, rather than just describe or predict. Note that it is possible to predict events or behaviors using a set of predictors, without necessarily explaining why such events are taking place. For instance, market analysts predict fluctuations in the stock market based on market announcements, earnings reports of major companies, and new data from the Federal Reserve and other agencies, based on previously observed correlations. Prediction requires only correlations. In contrast, explanations require causations, or understanding of cause-effect relationships. Establishing causation requires three conditions: (1) correlations between two constructs, (2) temporal precedence (the cause must precede the effect in time), and (3) rejection of alternative hypotheses (through testing). Scientific theories are different from theological, philosophical, or other explanations in that scientific theories can be empirically tested using scientific methods

Explanations can be idiographic or nomothetic. **Idiographic explanations** are those that explain a single situation or event in idiosyncratic detail. For example, you did poorly on an exam because: (1) you forgot that you had an exam on that day, (2) you arrived late to the exam due to a traffic jam, (3) you panicked midway through the exam, (4) you had to work late the previous evening and could not study for the exam, or even (5) your dog ate your textbook. The explanations may be detailed, accurate, and valid, but they may not apply to other similar situations, even involving the same person, and are hence not generalizable. In contrast, **nomothetic explanations** seek to explain a class of situations or events rather than a specific situation or event. For example, students who do poorly in exams do so because they did not spend adequate time preparing for exams or that they suffer from nervousness, attention deficit, or some other medical disorder. Because nomothetic explanations are designed to be generalizable across situations, events, or people, they tend to be less precise, less complete, and less detailed. However, they explain economically, using only a few explanatory variables. Because theories are also intended to serve as generalized explanations for patterns of events, behaviors, or phenomena, theoretical explanations are generally nomothetic in nature.

This chapter is reproduced with small alterations from from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at

https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

While understanding theories, it is also important to understand what theory is not. Theory is not data, facts, typologies, taxonomies, or empirical findings. A collection of facts is not a theory, just as a pile of stones is not a house. Likewise, a collection of constructs (e.g., a typology of constructs) is not a theory, because theories must go well beyond constructs to include propositions, explanations, and boundary conditions. Data, facts, and findings operate at the empirical or observational level, while theories operate at a conceptual level and are based on logic rather than observations.

There are many benefits to using theories in research. First, theories provide the underlying logic of the occurrence of natural or social phenomenon by explaining what are the key drivers and key outcomes of the target phenomenon and why, and what underlying processes are responsible driving that phenomenon. Second, they aid in sense-making by helping us synthesize prior empirical findings within a theoretical framework and reconcile contradictory findings by discovering contingent factors influencing the relationship between two constructs in different studies. Third, theories provide guidance for future research by helping identify constructs and relationships that are worthy of further research. Fourth, theories can contribute to cumulative knowledge building by bridging gaps between other theories and by causing existing theories to be reevaluated in a new light.

However, theories can also have their own share of limitations. As simplified explanations of reality, theories may not always provide adequate explanations of the phenomenon of interest based on a limited set of constructs and relationships. Theories are designed to be simple and parsimonious explanations, while reality may be significantly more complex. Furthermore, theories may impose blinders or limit researchers' "range of vision," causing them to miss out on important concepts that are not defined by the theory.

Building blocks of a theory

David Whetten (1989) suggests that there are four building blocks of a theory: constructs, propositions, logic, and boundary conditions/assumptions. Constructs capture the "what" of theories (i.e., what concepts are important for explaining a phenomenon), propositions capture the "how" (i.e., how are these concepts related to each other), logic represents the "why" (i.e., why are these concepts related), and boundary conditions/assumptions examines the "who, when, and where" (i.e., under what circumstances will these concepts and relationships work). Though constructs and propositions were previously discussed in Chapter 2, we describe them again here for the sake of completeness.

Constructs are abstract concepts specified at a high level of abstraction that are chosen specifically to explain the phenomenon of interest. Recall from Chapter 2 that constructs may be unidimensional (i.e., embody a single concept), such as weight or age, or multi-dimensional (i.e., embody multiple underlying concepts), such as personality or culture. While some constructs, such as age, education, and firm size, are easy to understand, others, such as creativity, prejudice, and organizational agility, may be more complex and abstruse, and still others such as trust, attitude, and learning, may represent temporal tendencies rather than

steady states. Nevertheless, all constructs must have clear and unambiguous operational definition that should specify exactly how the construct will be measured and at what level of analysis (individual, group, organizational, etc.). Measurable representations of abstract constructs are called **variables**. For instance, intelligence quotient (IQ score) is a variable that is purported to measure an abstract construct called intelligence. As noted earlier, scientific research proceeds along two planes: a theoretical plane and an empirical plane. Constructs are conceptualized at the theoretical plane, while variables are operationalized and measured at the empirical (observational) plane. Furthermore, variables may be independent, dependent, mediating, or moderating. The distinction between constructs (conceptualized at the theoretical level) and variables (measured at the empirical level) is shown in Figure 4.1.

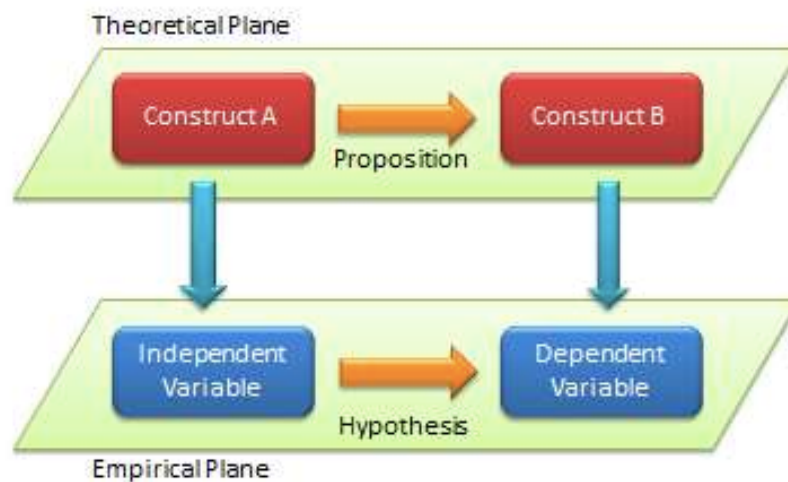


Figure 3.1 Distinction between theoretical and empirical concepts (Anol Bhattacharjee/[Social Science Research](#)).

Propositions are associations postulated between constructs based on deductive logic. Propositions are stated in declarative form and should ideally indicate a cause-effect relationship (e.g., if X occurs, then Y will follow). Note that propositions may be conjectural but **MUST** be testable, and should be rejected if they are not supported by empirical observations. However, like constructs, propositions are stated at the theoretical level, and they can only be tested by examining the corresponding relationship between measurable variables of those constructs. The empirical formulation of propositions, stated as relationships between variables, is called **hypotheses**. The distinction between propositions (formulated at the theoretical level) and hypotheses (tested at the empirical level) is depicted in Figure 4.1.

The third building block of a theory is the **logic** that provides the basis for justifying the propositions as postulated. Logic acts like a “glue” that connects the theoretical constructs and provides meaning and relevance to the relationships between these constructs. Logic also represents the “explanation” that lies at the core of a theory. Without logic, propositions will

be ad hoc, arbitrary, and meaningless, and cannot be tied into a cohesive “system of propositions” that is the heart of any theory.

Finally, all theories are constrained by assumptions about values, time, and space, and **boundary conditions** that govern where the theory can be applied and where it cannot be applied. For example, many economic theories assume that human beings are rational (or boundedly rational) and employ utility maximization based on cost and benefit expectations as a way of understand human behavior. In contrast, political science theories assume that people are more political than rational, and try to position themselves in their professional or personal environment in a way that maximizes their power and control over others. Given the nature of their underlying assumptions, economic and political theories are not directly comparable, and researchers should not use economic theories if their objective is to understand the power structure or its evolution in a organization. Likewise, theories may have implicit cultural assumptions (e.g., whether they apply to individualistic or collective cultures), temporal assumptions (e.g., whether they apply to early stages or later stages of human behavior), and spatial assumptions (e.g., whether they apply to certain localities but not to others). If a theory is to be properly used or tested, all of its implicit assumptions that form the boundaries of that theory must be properly understood. Unfortunately, theorists rarely state their implicit assumptions clearly, which leads to frequent misapplications of theories to problem situations in research.

Attributes of a good theory

Theories are simplified and often partial explanations of complex social reality. As such, there can be good explanations or poor explanations, and consequently, there can be good theories or poor theories. How can we evaluate the “goodness” of a given theory? Different criteria have been proposed by different researchers, the more important of which are listed below:

- **Logical consistency:** Are the theoretical constructs, propositions, boundary conditions, and assumptions logically consistent with each other? If some of these “building blocks” of a theory are inconsistent with each other (e.g., a theory assumes rationality, but some constructs represent non-rational concepts), then the theory is a poor theory.
- **Explanatory power:** How much does a given theory explain (or predict) reality? Good theories obviously explain the target phenomenon better than rival theories, as often measured by variance explained (R-square) value in regression equations.
- **Falsifiability:** British philosopher Karl Popper stated in the 1940’s that for theories to be valid, they must be falsifiable. Falsifiability ensures that the theory is potentially disprovable, if empirical data does not match with theoretical propositions, which allows for their empirical testing by researchers. In other words, theories cannot be theories unless they can be empirically testable. Tautological statements, such as “a day with high temperatures is a hot day” are not empirically testable because a hot day is defined (and measured) as a day with high temperatures, and hence, such statements cannot be viewed as a theoretical proposition. Falsifiability requires presence of rival explanations it ensures that the

This chapter is reproduced with small alterations from from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at

https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

constructs are adequately measurable, and so forth. However, note that saying that a theory is falsifiable is not the same as saying that a theory should be falsified. If a theory is indeed falsified based on empirical evidence, then it was probably a poor theory to begin with!

- **Parsimony:** Parsimony examines how much of a phenomenon is explained with how few variables. The concept is attributed to 14th century English logician Father William of Ockham (and hence called “Ockham’s razor” or “Occam’s razor”), which states that among competing explanations that sufficiently explain the observed evidence, the simplest theory (i.e., one that uses the smallest number of variables or makes the fewest assumptions) is the best. Explanation of a complex social phenomenon can always be increased by adding more and more constructs. However, such an approach defeats the purpose of having a theory, which is intended to be “simplified” and generalizable explanations of reality. Parsimony relates to the degrees of freedom in a given theory. Parsimonious theories have higher degrees of freedom, which allow them to be more easily generalized to other contexts, settings, and populations.

Approaches to theorizing

How do researchers build theories? Steinfeld and Fulk (1990) recommend four such approaches. The first approach is to build theories inductively based on observed patterns of events or behaviors. Such an approach is often called “grounded theory building”, because the theory is grounded in empirical observations. This technique is heavily dependent on the observational and interpretive abilities of the researcher, and the resulting theory may be subjective and non-confirmable. Furthermore, observing certain patterns of events will not necessarily make a theory, unless the researcher is able to provide consistent explanations for the observed patterns. We will discuss the grounded theory approach in a later chapter on qualitative research.

The second approach to theory building is to conduct a bottom-up conceptual analysis to identify different sets of predictors relevant to the phenomenon of interest using a predefined framework. One such framework may be a simple input-process-output framework, where the researcher may look for different categories of inputs, such as individual, organizational, and/or technological factors potentially related to the phenomenon of interest (the output), and describe the underlying processes that link these factors to the target phenomenon. This is also an inductive approach that relies heavily on the inductive abilities of the researcher, and interpretation may be biased by the researcher’s prior knowledge of the phenomenon being studied.

The third approach to theorizing is to extend or modify existing theories to explain a new context, such as by extending theories of individual learning to explain organizational learning. While making such an extension, certain concepts, propositions, and/or boundary conditions of the old theory may be retained and others modified to fit the new context. This

deductive approach leverages the rich inventory of social science theories developed by prior theoreticians, and is an efficient way of building new theories by building on existing ones.

The fourth approach is to apply existing theories in entirely new contexts by drawing upon the structural similarities between the two contexts. This approach relies on reasoning by analogy, and is probably the most creative way of theorizing using a deductive approach. For instance, Markus (1987) used analogic similarities between a nuclear explosion and uncontrolled growth of networks or network-based businesses to propose a critical mass theory of network growth. Just as a nuclear explosion requires a critical mass of radioactive material to sustain a nuclear explosion, Markus suggested that a network requires a critical mass of users to sustain its growth, and without such critical mass, users may leave the network, causing an eventual demise of the network.

Examples of social science theories

In this section, we present brief overviews of a few illustrative theories from different social science disciplines. These theories explain different types of social behaviors, using a set of constructs, propositions, boundary conditions, assumptions, and underlying logic. Note that the following represents just a simplistic introduction to these theories; readers are advised to consult the original sources of these theories for more details and insights on each theory.

Agency Theory. Agency theory (also called principal-agent theory), a classic theory in the organizational economics literature, was originally proposed by Ross (1973) to explain two-party relationships (such as those between an employer and its employees, between organizational executives and shareholders, and between buyers and sellers) whose goals are not congruent with each other. The goal of agency theory is to specify optimal contracts and the conditions under which such contracts may help minimize the effect of goal incongruence. The core assumptions of this theory are that human beings are self-interested individuals, boundedly rational, and risk-averse, and the theory can be applied at the individual or organizational level.

The two parties in this theory are the principal and the agent; the principal employs the agent to perform certain tasks on its behalf. While the principal's goal is quick and effective completion of the assigned task, the agent's goal may be working at its own pace, avoiding risks, and seeking self-interest (such as personal pay) over corporate interests. Hence, the goal incongruence occurs. Compounding the nature of the problem may be information asymmetry problems caused by the principal's inability to adequately observe the agent's behavior or accurately evaluate the agent's skill sets. Such asymmetry may lead to agency problems where the agent may not put forth the effort needed to get the task done (the moral hazard problem) or may misrepresent its expertise or skills to get the job but not perform as expected (the adverse selection problem). Typical contracts that are behavior-based, such as a monthly salary, cannot overcome these problems. Hence, agency theory recommends using outcome-based contracts, such as commissions or a fee payable upon task completion, or mixed contracts that combine behavior-based and outcome-based incentives.

This chapter is reproduced with small alterations from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

An employee stock option plan is an example of an outcome-based contract while employee pay is a behavior-based contract. Agency theory also recommends tools that principals may employ to improve the efficacy of behavior-based contracts, such as investing in monitoring mechanisms (such as hiring supervisors) to counter the information asymmetry caused by moral hazard, designing renewable contracts contingent on agent's performance (performance assessment makes the contract partially outcome-based), or by improving the structure of the assigned task to make it more programmable and therefore more observable.

Theory of Planned Behavior. Postulated by Azjen (1991), the theory of planned behavior (TPB) is a generalized theory of human behavior in the social psychology literature that can be used to study a wide range of individual behaviors. It presumes that individual behavior represents conscious reasoned choice, and is shaped by cognitive thinking and social pressures. The theory postulates that behaviors are based on one's intention regarding that behavior, which in turn is a function of the person's attitude toward the behavior, subjective norm regarding that behavior, and perception of control over that behavior (see Figure 4.2). Attitude is defined as the individual's overall positive or negative feelings about performing the behavior in question, which may be assessed as a summation of one's beliefs regarding the different consequences of that behavior, weighted by the desirability of those consequences. Subjective norm refers to one's perception of whether people important to that person expect the person to perform the intended behavior, and represented as a weighted combination of the expected norms of different referent groups such as friends, colleagues, or supervisors at work. Behavioral control is one's perception of internal or external controls constraining the behavior in question. Internal controls may include the person's ability to perform the intended behavior (self-efficacy), while external control refers to the availability of external resources needed to perform that behavior (facilitating conditions). TPB also suggests that sometimes people may intend to perform a given behavior but lack the resources needed to do so, and therefore posits that behavioral control can have a direct effect on behavior, in addition to the indirect effect mediated by intention.



Figure 3.2 Theory of planned behaviour (Anol Bhattacharjee/[Social Science Research](https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks)).

TPB is an extension of an earlier theory called the theory of reasoned action, which included attitude and subjective norm as key drivers of intention, but not behavioral control. The latter construct was added by Ajzen in TPB to account for circumstances when people may have incomplete control over their own behaviors (such as not having high-speed Internet access for web surfing).

Innovation diffusion theory. Innovation diffusion theory (IDT) is a seminal theory in the communications literature that explains how innovations are adopted within a population of potential adopters. The concept was first studied by French sociologist Gabriel Tarde, but the theory was developed by Everett Rogers in 1962 based on observations of 508 diffusion studies. The four key elements in this theory are: innovation, communication channels, time, and social system. Innovations may include new technologies, new practices, or new ideas, and adopters may be individuals or organizations. At the macro (population) level, IDT views innovation diffusion as a process of communication where people in a social system learn about a new innovation and its potential benefits through communication channels (such as mass media or prior adopters) and are persuaded to adopt it. Diffusion is a temporal process; the diffusion process starts off slow among a few early adopters, then picks up speed as the innovation is adopted by the mainstream population, and finally slows down as the adopter population reaches saturation. The cumulative adoption pattern therefore an S-shaped curve, as shown in Figure 4.3, and the adopter distribution represents a normal distribution. All adopters are not identical, and adopters can be classified into innovators, early adopters, early majority, late majority, and laggards based on their time of their adoption. The rate of diffusion also depends on characteristics of the social system such as the presence of opinion leaders (experts whose opinions are valued by others) and change agents (people who influence others' behaviors).

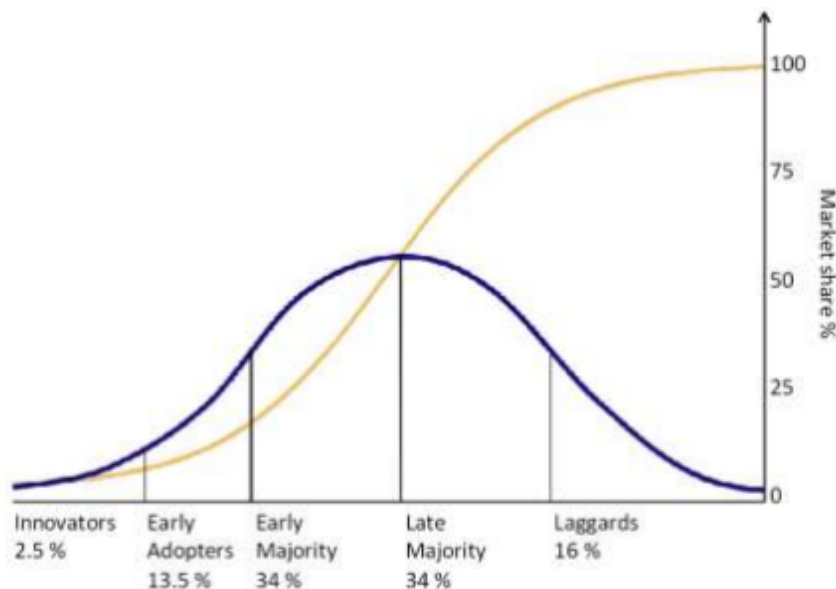


Figure 3.3 S-shaped diffusion curve (Anol Bhattacharjee/[Social Science Research](#)).

At the micro (adopter) level, Rogers (1995) suggests that innovation adoption is a process consisting of five stages: (1) knowledge: when adopters first learn about an innovation from mass-media or interpersonal channels, (2) persuasion: when they are persuaded by prior adopters to try the innovation, (3) decision: their decision to accept or reject the innovation, (4) implementation: their initial utilization of the innovation, and (5) confirmation: their decision to continue using it to its fullest potential (see Figure 4.4). Five innovation characteristics are presumed to shape adopters' innovation adoption decisions: (1) relative advantage: the expected benefits of an innovation relative to prior innovations, (2) compatibility: the extent to which the innovation fits with the adopter's work habits, beliefs, and values, (3) complexity: the extent to which the innovation is difficult to learn and use, (4) trialability: the extent to which the innovation can be tested on a trial basis, and (5) observability: the extent to which the results of using the innovation can be clearly observed. The last two characteristics have since been dropped from many innovation studies. Complexity is negatively correlated to innovation adoption, while the other four factors are positively correlated. Innovation adoption also depends on personal factors such as the adopter's risk-taking propensity, education level, cosmopolitanism, and communication influence. Early adopters are venturesome, well educated, and rely more on mass media for information about the innovation, while later adopters rely more on interpersonal sources (such as friends and family) as their primary source of information. IDT has been criticized for having a "pro-innovation bias," that is for presuming that all innovations are beneficial and will be eventually diffused across the entire population, and because it does not allow for inefficient innovations such as fads or fashions to die off quickly without being adopted by the entire population or being replaced by better innovations.

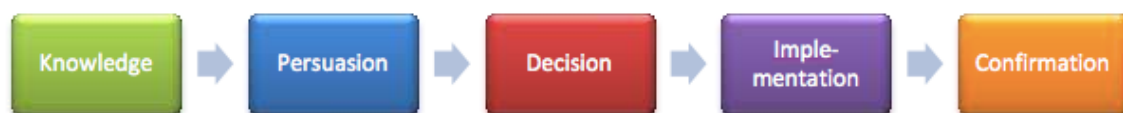


Figure 3.4 Innovation adoption process (Anol Bhattacharjee/[Social Science Research](#)).

Elaboration Likelihood Model. Developed by Petty and Cacioppo (1986), the elaboration likelihood model (ELM) is a dual-process theory of attitude formation or change in the psychology literature. It explains how individuals can be influenced to change their attitude toward a certain object, events, or behavior and the relative efficacy of such change strategies. The ELM posits that one's attitude may be shaped by two "routes" of influence, the central route and the peripheral route, which differ in the amount of thoughtful information processing or "elaboration" required of people (see Figure 4.5). The central route requires a person to think about issue-related arguments in an informational message and carefully scrutinize the merits and relevance of those arguments, before forming an informed judgment about the target object. In the peripheral route, subjects rely on external "cues" such as number of prior users, endorsements from experts, or likeability of the endorser, rather than on the quality of arguments, in framing their attitude towards the target object.

This chapter is reproduced with small alterations from from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

The latter route is less cognitively demanding, and the routes of attitude change are typically operationalized in the ELM using the *argument quality* and *peripheral cues* constructs respectively.

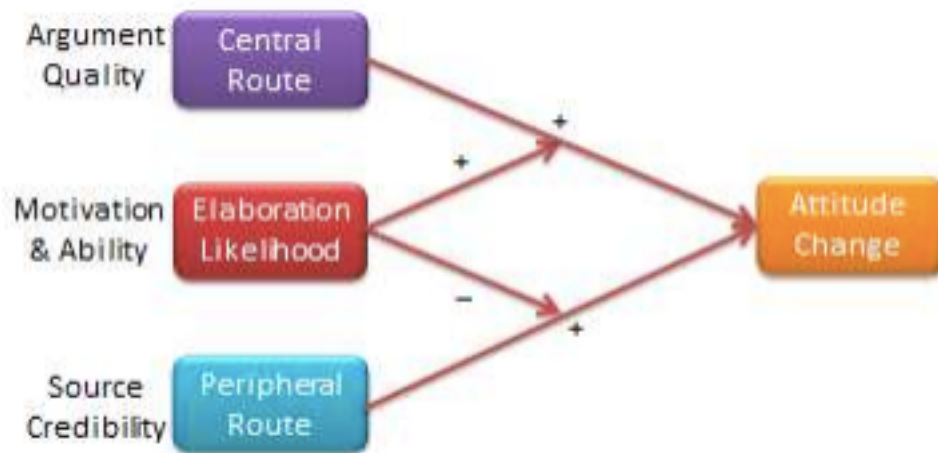


Figure 3.5 Elaboration likelihood model (Anol Bhattacharjee/[Social Science Research](#)).

Whether people will be influenced by the central or peripheral routes depends upon their ability and motivation to elaborate the central merits of an argument. This ability and motivation to elaborate is called *elaboration likelihood*. People in a state of high elaboration likelihood (high ability and high motivation) are more likely to thoughtfully process the information presented and are therefore more influenced by argument quality, while those in the low elaboration likelihood state are more motivated by peripheral cues. Elaboration likelihood is a situational characteristic and not a personal trait. For instance, a doctor may employ the central route for diagnosing and treating a medical ailment (by virtue of his or her expertise of the subject), but may rely on peripheral cues from auto mechanics to understand the problems with his car. As such, the theory has widespread implications about how to enact attitude change toward new products or ideas and even social change.

General Deterrence Theory. Two utilitarian philosophers of the eighteenth century, Cesare Beccaria and Jeremy Bentham, formulated General Deterrence Theory (GDT) as both an explanation of crime and a method for reducing it. GDT examines why certain individuals engage in deviant, anti-social, or criminal behaviors. This theory holds that people are fundamentally rational (for both conforming and deviant behaviors), and that they freely choose deviant behaviors based on a rational cost-benefit calculation. Because people naturally choose utility-maximizing behaviors, deviant choices that engender personal gain or pleasure can be controlled by increasing the costs of such behaviors in the form of punishments (countermeasures) as well as increasing the probability of apprehension. Swift, severity, and certainty of punishments are the key constructs in GDT.

While classical positivist research in criminology seeks generalized causes of criminal behaviors, such as poverty, lack of education, psychological conditions, and recommends

strategies to rehabilitate criminals, such as by providing them job training and medical treatment, GDT focuses on the criminal decision-making process and situational factors that influence that process. Hence, a criminal's personal situation (such as his personal values, his affluence, and his need for money) and the environmental context (such as how protected is the target, how efficient is the local police, how likely are criminals to be apprehended) play key roles in this decision-making process. The focus of GDT is not how to rehabilitate criminals and avert future criminal behaviors, but how to make criminal activities less attractive and therefore prevent crimes. To that end, "target hardening" such as installing deadbolts and building self-defense skills, legal deterrents such as eliminating parole for certain crimes, "three strikes law" (mandatory incarceration for three offenses, even if the offenses are minor and not worth imprisonment), and the death penalty, increasing the chances of apprehension using means such as neighborhood watch programs, special task forces on drugs or gang-related crimes, and increased police patrols, and educational programs such as highly visible notices such as "Trespassers will be prosecuted" are effective in preventing crimes. This theory has interesting implications not only for traditional crimes, but also for contemporary white-collar crimes such as insider trading, software piracy, and illegal sharing of music.

This chapter is reproduced with small alterations from from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

References chapter 3

Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes* (50), 179-211.

Bacharach, S.B. (1989). Organizational theories: some criteria for evaluation. *Academy of Management Review*, 14(4), 496-515.

Markus, M.L. (1987). Toward a 'critical mass' theory of interactive media: universal access, interdependence, and diffusion. *Communication Research*, 14(5), 491-511.

Petty, R.E. & Cacioppo, J.T. (1986). *Communication and Persuasion: Central and Peripheral Routes to Attitude Change*. New York: Springer-Verlag.

Rogers, E. (1962). *Diffusion of Innovations*. New York: The Free Press. Other editions 1983, 1996, 2005.

Ross, S.A. (1973). The economic theory of agency: the principal's problem. *American Economic Review*, 63(2), 134-139.

Steinfeld, C.W. & Fulk, J. (1990). The Theory Imperative. In J. Fulk & C.W. Steinfeld (Eds.), *Organizations and Communications Technology*. Newbury Park, CA: Sage Publications.

This chapter is reproduced with small alterations from from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

Chapter 4: Research ethics

Ethics is defined by Webster's dictionary as conformance to the standards of conduct of a given profession or group. Such standards are often defined at a disciplinary level through a professional code of conduct, and sometimes enforced by university committees called the Institutional Review Board. Even if not explicitly specified, scientists are still expected to be aware of and abide by general agreements shared by the scientific community on what constitutes acceptable and unacceptable behaviors in the professional conduct of science. For instance, scientists should not manipulate their data collection, analysis, and interpretation procedures in a way that contradicts the principles of science or the scientific method or advances their personal agenda.

Why is research ethics important? Because, science has often been manipulated in unethical ways by people and organizations to advance their private agenda and engage in activities that are contrary to the norms of scientific conduct. A classic example is pharmaceutical giant Merck's drug trials of Vioxx, where the company hid the fatal side-effects of the drug from the scientific community, resulting in 3468 deaths of Vioxx recipients, mostly from cardiac arrest. In 2010, the company agreed to a \$4.85 billion settlement and appointed two independent committees and a chief medical officer to monitor the safety of its drug development process. Merck's conduct was unethical and violated the scientific principles of data collection, analysis, and interpretation.

Ethics is the moral distinction between right and wrong, and what is unethical may not necessarily be illegal. If a scientist's conduct falls within the gray zone between ethics and law, she may not be culpable in the eyes of the law, but may still be ostracized in her professional community, face severe damage to professional reputation, and may even lose her job on grounds of professional misconduct. These ethical norms may vary from one society to another, and here, we refer to ethical standards as applied to scientific research in Western countries.

4.1 Ethical principles in scientific research

Some of the expected tenets of ethical behavior that are widely accepted within the scientific community are as follows.

Voluntary participation and harmlessness. Subjects in a research project must be aware that their participation in the study is voluntary, that they have the freedom to withdraw from the study at any time without any unfavorable consequences, and they are not harmed as a result of their participation or non-participation in the project. The most flagrant violations of the voluntary participation principle are probably forced medical experiments conducted by Nazi researchers on prisoners of war during World War II, as documented in the post-War Nuremberg Trials (these experiments also originated the term "crimes against humanity"). Less known violations include the Tuskegee syphilis experiments conducted by the U.S. Public Health Service during 1932-1972, in which nearly 400 impoverished African-American men

This chapter is reproduced with small alterations from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

suffering from syphilis were denied treatment even after penicillin was accepted as an effective treatment of syphilis, and subjects were presented with false treatments such as spinal taps as cures for syphilis. Even if subjects face no mortal threat, they should not be subjected to personal agony as a result of their participation. In 1971, psychologist Philip Zimbardo created the Stanford Prison Experiment, where Stanford students recruited as subjects were randomly assigned to roles such as prisoners or guards. When it became evident that student prisoners were suffering psychological damage as a result of their mock incarceration and student guards were exhibiting sadism that would later challenge their own self-image, the experiment was terminated.

Today, if an instructor asks her students to fill out a questionnaire and informs them that their participation is voluntary, students must not fear that their non-participation may hurt their grade in class in any way. For instance, it is unethical to provide bonus points for participation and no bonus points for non-participations, because it places non-participants at a distinct disadvantage. To avoid such circumstances, the instructor may possibly provide an alternate task for non-participants so that they can recoup the bonus points without participating in the research study, or by providing bonus points to everyone irrespective of their participation or non-participation. Furthermore, all participants must receive and sign an **Informed Consent** form that clearly describes their right to not participate and right to withdraw, before their responses in the study can be recorded. In a medical study, this form must also specify any possible risks to subjects from their participation. For subjects under the age of 18, this form must be signed by their parent or legal guardian. Researchers must retain these informed consent forms for a period of time (often three years) after the completion of the data collection process in order to comply with the norms of scientific conduct in their discipline or workplace.

Anonymity and confidentiality. To protect subjects' interests and future well-being, their identity must be protected in a scientific study. This is done using the dual principles of anonymity and confidentiality. **Anonymity** implies that the researcher or readers of the final research report or paper cannot identify a given response with a specific respondent. An example of anonymity in scientific research is a mail survey in which no identification numbers are used to track who is responding to the survey and who is not. In studies of deviant or undesirable behaviors, such as drug use or illegal music downloading by students, truthful responses may not be obtained if subjects are not assured of anonymity. Further, anonymity assures that subjects are insulated from law enforcement or other authorities who may have an interest in identifying and tracking such subjects in the future.

In some research designs such as face-to-face interviews, anonymity is not possible. In other designs, such as a longitudinal field survey, anonymity is not desirable because it prevents the researcher from matching responses from the same subject at different points in time for longitudinal analysis. Under such circumstances, subjects should be guaranteed **confidentiality**, in which the researcher can identify a person's responses, but promises not to divulge that person's identity in any report, paper, or public forum. Confidentiality is a

weaker form of protection than anonymity, because social research data do not enjoy the “privileged communication” status in United State courts as do communication with priests or lawyers. For instance, two years after the Exxon Valdez supertanker spilled ten million barrels of crude oil near the port of Valdez in Alaska, the communities suffering economic and environmental damage commissioned a San Diego research firm to survey the affected households about personal and embarrassing details about increased psychological problems in their family. Because the cultural norms of many Native Americans made such public revelations particularly painful and difficult, respondents were assured confidentiality of their responses. When this evidence was presented to court, Exxon petitioned the court to subpoena the original survey questionnaires (with identifying information) in order to cross-examine respondents regarding their answers that they had given to interviewers under the protection of confidentiality, and was granted that request. Luckily, the Exxon Valdez case was settled before the victims were forced to testify in open court, but the potential for similar violations of confidentiality still remains.

In one extreme case, Rick Scarce, a graduate student at Washington State University, conducted participant observation studies of animal rights activists, and chronicled his findings in a 1990 book called *Ecowarriors: Understanding the Radical Environmental Movement*. In 1993, Scarce was called before a grand jury to identify the activists he studied. The researcher refused to answer grand jury questions, in keeping with his ethical obligations as a member of the American Sociological Association, and was forced to spend 159 days at Spokane County Jail. To protect themselves from travails similar to Rick Scarce, researchers should remove any identifying information from documents and data files as soon as they are no longer necessary. In 2002, the United States Department of Health and Human Services issued a “Certificate of Confidentiality” to protect participants in research projects from police and other authorities. Not all research projects qualify for this protection, but this can provide an important support for protecting participant confidentiality in many cases.

Disclosure. Usually, researchers have an obligation to provide some information about their study to potential subjects before data collection to help them decide whether or not they wish to participate in the study. For instance, who is conducting the study, for what purpose, what outcomes are expected, and who will benefit from the results. However, in some cases, disclosing such information may potentially bias subjects’ responses. For instance, if the purpose of a study is to examine to what extent subjects will abandon their own views to conform with “groupthink” and they participate in an experiment where they listen to others’ opinions on a topic before voicing their own, then disclosing the study’s purpose before the experiment will likely sensitize subjects to the treatment. Under such circumstances, even if the study’s purpose cannot be revealed before the study, it should be revealed in a debriefing session immediately following the data collection process, with a list of potential risks or harm borne by the participant during the experiment.

Analysis and reporting. Researchers also have ethical obligations to the scientific community on how data is analyzed and reported in their study. Unexpected or negative findings should

This chapter is reproduced with small alterations from from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

be fully disclosed, even if they cast some doubt on the research design or the findings. Similarly, many interesting relationships are discovered after a study is completed, by chance or data mining. It is unethical to present such findings as the product of deliberate design. In other words, hypotheses should not be designed in positivist research after the fact based on the results of data analysis, because the role of data in such research is to test hypotheses, and not build them. It is also unethical to “carve” their data into different segments to prove or disprove their hypotheses of interest, or to generate multiple papers claiming different data sets. Misrepresenting questionable claims as valid based on partial, incomplete, or improper data analysis is also dishonest. Science progresses through openness and honesty, and researchers can best serve science and the scientific community by fully disclosing the problems with their research, so that they can save other researchers from similar problems.

4.2 Institutional review boards

Research ethics in studies involving human subjects is governed in the United States by federal law. Any agency, such as a university or a hospital, that wants to apply for federal funding to support its research projects must establish that it is in compliance with federal laws governing the rights and protection of human subjects. This process is overseen by a panel of experts in that agency called an **Institutional Review Board (IRB)**. The IRB reviews all research proposals involving human subjects to ensure that the principles of voluntary participation, harmlessness, anonymity, confidentiality, and so forth are preserved, and that the risks posed to human subjects are minimal. Even though the federal laws apply specifically for federally funded projects, the same standards and procedures are also applied to non-funded or even student projects.

The IRB approval process requires completing a structured application providing complete information about the research project, the researchers (principal investigators), and details on how the subjects’ rights will be protected. Additional documentation such as the Informed Consent form, research questionnaire or interview protocol may be needed. The researchers must also demonstrate that they are familiar with the principles of ethical research by providing certification of their participation in an research ethics course. Data collection can commence only after the project is cleared by the IRB review committee.

Professional code of ethics

Most professional associations of researchers have established and published formal codes of conduct describing what constitute acceptable and unacceptable professional behavior of their member researchers. As an example, the summarized code of conduct for the Association of Information Systems (AIS), the global professional association of researchers in the information systems discipline, is summarized in Table 16.1 (the complete code of conduct is available [online](#)). Similar codes of ethics are also available for other disciplines.

The AIS code of conduct groups ethical violations in two categories. Category I includes serious transgressions such as plagiarism and falsification of data, research procedures, or data analysis, which may lead to expulsion from the association, dismissal from employment,

This chapter is reproduced with small alterations from from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

legal action, and fatal damage to professional reputation. Category 2 includes less serious transgression such as not respecting the rights of research subjects, misrepresenting the originality of research projects, and using data published by others without acknowledgement, which may lead to damage to professional reputation, sanctions from journals, and so forth. The code also provides guidance on good research behaviors, what to do when ethical transgressions are detected (for both the transgressor and the victim), and the process to be followed by AIS in dealing with ethical violation cases. Though codes of ethics such as this have not completely eliminated unethical behavior, they have certainly helped clarify the boundaries of ethical behavior in the scientific community and reduced instances of ethical transgressions.

Table 4.1 Code of ethics for the Association of Information Systems

CATEGORY ONE: Codes in this category must ALWAYS be adhered to and disregard for them constitutes a serious ethical breach. Serious breaches can result in your expulsion from academic associations, dismissal from your employment, legal action against you, and potentially fatal damage to your academic reputation. 1. Do not plagiarize. 2. Do not fabricate or falsify data, research procedures, or data analysis.
CATEGORY TWO: Codes in this category are recommended ethical behavior. Flagrant disregard of these or other kinds of professional etiquette, while less serious, can result in damage to your reputation, editorial sanctions, professional embarrassment, legal action, and the ill will of your colleagues. 1. Respect the rights of research subjects, particularly their rights to information privacy, and to being informed about the nature of the research and the types of activities in which they will be asked to engage. 2. Do not make misrepresentations to editors and conference program chairs about the originality of papers you submit to them. 3. Do not abuse the authority and responsibility you have been given as an editor, reviewer or supervisor, and ensure that personal relationships do not interfere with your judgement. 4. Declare any material conflict of interest that might interfere with your ability to be objective and impartial when reviewing submissions, grant applications, software, or undertaking work from outside sources. 5. Do not take or use published data of others without acknowledgement, or unpublished data without both permission and acknowledgement. 6. Acknowledge the substantive contributions of all research participants, whether colleagues or students, according to their intellectual contribution. 7. Do not use other people's unpublished writings, information, ideas, concepts or data that you may see as a result of processes such as peer review without permission of the author. 8. Use archival material only in accordance with the rules of the archival source.

ADVICE: Some suggestions on how to protect yourself from authorship disputes, mis-steps, mistakes, and even legal action.

1. Keep the documentation and data necessary to validate your original authorship for each scholarly work with which you are connected.
2. Do not republish old ideas of your own as if they were a new intellectual contribution.
3. Settle data set ownership issues before data compilation.
4. Consult appropriate colleagues if in doubt.

An ethical controversy

Robert Allen “Laud” Humphreys is an American sociologist and author, who is best known for his Ph.D. dissertation, *Tearoom Trade*, published in 1970. This book is an ethnographic account of anonymous male homosexual encounters in public toilets in parks – a practice known as “tea-rooming” in U.S. gay slang. Humphreys was intrigued by the fact that the majority of participants in tearoom activities were outwardly heterosexual men, who lived otherwise conventional family lives in their communities. However, it was important to them to preserve their anonymity during tearoom visits.

Typically, tearoom encounters involved three people – the two males engaging in a sexual act and a lookout person called a “watchqueen.” The job of the watchqueen was to alert the two participating males for police or other people, while deriving pleasure from watching the action as a voyeur. Because it was not otherwise possible to reach these subjects, Humphreys showed up at public toilets, masquerading as a watchqueen. As a participant observer, Humphreys was able to conduct field observations for his dissertation, as he normally would in a study of political protests or any other sociological phenomenon.

Humphreys needed more information on the participants. But because participants were unwilling to be interviewed in the field or disclose personal identities, Humphreys wrote down the license plate numbers of the participants’ cars, wherever possible, and tracked down their names and addresses from public databases. Then he visited these men at their homes, disguising himself to avoid recognition and announcing that he was conducting a survey, and collected personal data that was not otherwise available.

Humphreys’ research generated considerable controversy within the scientific community. Many critics said that he should not have invaded others’ right to privacy in the name of science, others were worried about his deceitful behavior in leading participants to believe that he was only a watchqueen, when he clearly had ulterior motives. Even those who considered observing tearoom activity to be acceptable because the participants used public facilities, thought that the follow-up interview survey in participants’ homes under false pretenses was unethical, because of the way he obtained their home addresses and because he did not seek informed consent. A few researchers justified Humphrey’s approach saying that this was an important sociological phenomenon worth investigating, that there was no other way to collect this data, and that the deceit was harmless, since Humphreys did not

This chapter is reproduced with small alterations from from **Social Science Research: Principles, Methods, and Practices** (2nd edition), written by Anol Bhattacharjee (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012), licensed under CC BY-NC-SA 3.0. Access for free at https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks

disclose his subjects' identities to anyone. This controversy was never resolved, and it is still hotly debated in classes and forums on research ethics.

References chapter 4

Haney, C., Banks, W.C. & Zimbardo, P.G. (1973). Interpersonal dynamics in a simulated prison. *International Journal of Criminology and Penology*, 1, 69–97.

Humphreys, L. (2017). *Tearoom trade: impersonal sex in public places*. Oxon: Routledge.

Scarce, R. (2016). *Eco-warriors: understanding the radical environmental movement*. New York: Routledge.

Attribution and acknowledgements

This work is a derivative of the following works:

1. Heather Griffiths and Nathan Keirns, *Introduction to Sociology (2e)* (Houston, TX: OpenStax, 2020). Available at <https://openstax.org/details/books/introduction-sociology-2e>. Licensed under a Creative Commons Attribution License v4.0. The original text has been visually edited and the numbering of sections has been adjusted.
2. Steven Graves, *Introduction to Human Geography: A Disciplinary Approach (3e)* (2020). Available at <https://sites.google.com/site/gravesgeography/home?authuser=0>. Published online and “instructors and students are welcome to use the text free of charge”. The original text has been visually edited and the numbering of sections has been adjusted.
3. Anol Bhattacharjee, *Social Science Research: Principles, Methods, and Practices (2e)* (Tampa, FL: USF Tampa Library Open Access Collections at Scholar Commons, 2012). Available at https://scholarcommons.usf.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1002&context=oa_textbooks. Licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 3.0 Unported license. The original text has been visually edited and the numbering of sections has been adjusted.

In this edition, only minor textual changes have been made for clarity and correct references.

This edition has been compiled by:

Sander van Lanen (University of Groningen, The Netherlands)

www.rug.nl/staff/s.van.lanen/ - <https://orcid.org/0000-0002-0877-4508>

The editor would like to thank Myroslava Zhuk, Frederiek van Rij, and Raquel Raj (University of Groningen) for their information, encouragement and advice on the compilation of this textbook.

January 2021

This work is licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International \(CC BY-NC-SA 4.0\)](https://creativecommons.org/licenses/by-nc-sa/4.0/).

